

The Mobility Frontier: Why the Intergenerational Elasticity Is an Insufficient Statistic

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Abstract

The intergenerational elasticity (IGE) has served as the workhorse measure of economic mobility for three decades. This paper asks what the IGE identifies about the structural transmission of status across generations, and proves the answer is: nothing uniquely. I model the life course as a sequence of K ordered gates—early childhood, schooling, tertiary access, labor market entry—each specified as a recursive ordered probit with percolation rates, bypass coefficients, and cutpoints. Six results follow. First, a *non-identification theorem*: the mapping from structural parameters to the IGE has pre-images of dimension $d - 1$, where d is the structural parameter count (Theorem 1). Second, the linear channel decomposition is nested as the continuous limit where cutpoints vanish (Lemma 1). Third, a *distributional multiplier*: the effect of gate-widening policies on the IGE depends on population mass at the cutpoint, an object invisible to linear models (Proposition 1). Fourth, *non-uniqueness*: two economies with the same IGE can respond in opposite directions to the same policy (Proposition 2). Fifth, a *Shapley decomposition* attributes the IGE to stages approximately in proportion to their Fisher information, with an explicit error bound (Proposition 4). Sixth, *optimal targeting* under convex costs equalizes the distributional-multiplier-adjusted return per dollar across stages, yielding interior solutions that can reverse the Heckman early-investment ordering (Proposition 5). The *mobility frontier*—the feasible set of child destinations conditional on origin—replaces the IGE as the sufficient object for policy analysis.

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1 Introduction

The intergenerational elasticity of income—the slope of child log-income on parent log-income—has organized empirical mobility research since Solon (1992) and Zimmerman (1992) revised the United States estimate upward from approximately 0.2 to 0.4. In the three decades since, refinements in measurement have been substantial: Mazumder (2005) demonstrated that even five-year income averages produce attenuation bias and estimated the IGE at approximately 0.6 using sixteen-year averages; Haider and Solon (2006) characterized lifecycle bias and established that measurement around age forty minimizes it; Nybom and Stuhler (2016, 2017) showed that lifecycle bias persists at roughly twenty percent even at optimal ages; and Chetty, Hendren, Kline, and Saez (2014) documented enormous geographic variation using administrative data on over forty million parent-child pairs. Yet throughout this period, the conceptual structure of the IGE—what it identifies about the mechanism of transmission—has gone largely uninterrogated. The IGE tells us how persistent a society is. It does not tell us *where* in the life course that persistence concentrates, *which* gates are binding, or *how* the same numerical persistence can arise from structurally distinct economies with opposite policy implications.

This paper proves that the IGE is structurally degenerate: it maps to an infinite-dimensional set of structural economies, so that knowing the IGE reveals essentially nothing about the life-course architecture generating it. Theorem 1 establishes that the pre-image of any IGE value under the structural model is a smooth manifold of dimension $d - 1$, where d counts the percolation rates, bypass coefficients, cutpoints, and return parameters. Proposition 2 shows this degeneracy has operational consequences: two economies with identical IGEs can respond in opposite directions to the same gate-widening policy. The linear channel decomposition of Bowles and Gintis (2002) and Blanden, Gregg, and Macmillan (2007) inherits the same deficiency, because Lemma 1 proves it is the continuous limit of the structural model—the special case in which gate locations play no role and the distributional multiplier (Proposition 1) vanishes identically.

The structural model replaces the IGE with a richer object: the *mobility frontier*, defined as the feasible set of child destinations conditional on parental origin. The life course is modeled as a sequence of K ordered gates—early childhood development, primary and secondary schooling, tertiary access, and labor market entry—each specified as a recursive ordered probit. At each gate, a latent index depends on the previous stage (the *percolation rate* γ_k) and on parental status directly (the *bypass coefficient* δ_k). The observed stage outcome is determined by where the latent index falls relative to *cutpoints* $\tau_{k,j}$ that locate the gate boundaries. The cutpoints are the new policy-relevant objects: they determine

the difficulty of each transition, and their location relative to the population distribution creates the distributional multiplier that governs the return to intervention. The Shapley decomposition distributes the IGE across stages with axiomatic guarantees (Proposition 4), and optimal targeting under convex costs equalizes the multiplier-adjusted return per dollar across stages (Proposition 5), yielding interior allocations that can reverse the standard Heckman early-investment ordering when early gates are already permissive but later gates are restrictive.

The paper engages three literatures. The IGE measurement literature, from the foundational contributions of Solon (1992, 1999) through the recent critiques of Mitnik and Grusky (2020), has progressively documented the fragility of the IGE as a summary statistic: it is sensitive to income averaging horizons, lifecycle measurement age, rank versus log transformations, and the distinction between elasticities of expected income versus conditional expectations. Theorem 1 shows that these are symptoms of a deeper problem—the IGE is structurally degenerate, not merely noisily estimated.

The channel decomposition literature has attempted to open the black box by decomposing the IGE into additive contributions of mediating variables. Bowles and Gintis (2002) decomposed the intergenerational correlation into IQ, schooling, wealth, and personality components but found a large unexplained residual. Blanden, Gregg, and Macmillan (2007) used UK cohort data to explain over eighty percent of the rise in persistence through cognitive skills, noncognitive traits, and education. Lefgren, Lindquist, and Sims (2012) separated causal-income from endowment components using Swedish data. These decompositions are valuable but carry two structural limitations. First, they are order-dependent and linear. Second—and less recognized—they require continuous mediators: the additive decomposition $\beta = \sum_k \alpha_k \gamma_k$ breaks down when intermediate outcomes are discrete (grade completion, occupational category, employment sector) because the variance normalization in nonlinear models creates a rescaling confound that Karlson, Holm, and Breen (2012) and Breen, Karlson, and Holm (2013) documented extensively. In practice, the life-course variables that matter most for mobility—whether a child completes secondary school, enters the formal sector, reaches a managerial tier—are inherently ordered and discrete. The percolation model accommodates this directly: the cutpoints $\tau_{k,j}$ define the discrete categories, and the number of categories J_k at each stage is a parameter the researcher chooses to match the data. Lemma 1 shows that the linear decomposition is the continuous limit $J_k \rightarrow \infty$ —the special case where the ordered structure vanishes and the distributional multiplier plays no role.

The Shapley decomposition literature, initiated in distributional analysis by Shorrocks (2013) and applied to R^2 decomposition by Israeli (2007), provides the axiomatic frame-

work for order-independent attribution. Brunori, Ferreira, and Neidhöfer (2025) applied Shapley to inequality of opportunity in Latin America—the closest existing exercise. The present paper deploys the Shapley value for a purpose it has never served: decomposing the IGE itself across life-course stages, with a proportionality approximation (Proposition 4) linking Shapley shares to Fisher information at each gate.

The paper proceeds as follows. Section 2 defines the percolation model, the reduced form, and the mobility frontier. Section 3 proves the non-identification theorem. Section 4 develops the policy consequences: the nesting lemma, the distributional multiplier, the non-uniqueness result, and the multi-gate extension. Section 5 presents the Shapley decomposition and optimal targeting. Section 6 discusses the Great Gatsby Curve, cutpoints as policy objects, and limitations. Section 7 locates the contribution relative to seven additional literatures. Section 8 concludes.

2 The Percolation Model

2.1 Setup

Let $S_0 \in \mathbb{R}$ denote parental status—log permanent income or a latent index of socioeconomic position. A child’s life course unfolds through K ordered stages $S_1, S_2, \dots, S_K$, each taking values in $\{0, 1, \dots, J_k\}$. The terminal outcome is adult income $Y \in \mathbb{R}$.

Definition 1 (Percolation model). The percolation model $\mathcal{P}(K, \{J_k\}, \gamma, \delta, \tau, \alpha, \delta_Y)$ is the recursive system

$$S_1^* = \gamma_1 S_0 + \varepsilon_1, \tag{1}$$

$$S_k^* = \gamma_k S_{k-1}^* + \delta_k S_0 + \varepsilon_k, \quad k = 2, \dots, K, \tag{2}$$

$$Y = \alpha S_K^* + \delta_Y S_0 + \varepsilon_Y, \tag{3}$$

where S_k^* is a latent index and the observed stage outcome is

$$S_k = j \iff \tau_{k,j-1} < S_k^* \leq \tau_{k,j}, \quad j = 0, 1, \dots, J_k, \tag{4}$$

with boundary conventions $\tau_{k,-1} = -\infty$ and $\tau_{k,J_k} = +\infty$. The errors $\varepsilon_1, \dots, \varepsilon_K, \varepsilon_Y$ are mutually independent standard normal, a normalization that identifies the variance scale in each ordered probit.

The parameters carry the following interpretation. The *percolation rate* γ_k governs how strongly the previous latent position propagates forward through stage k . The *bypass*

coefficient δ_k captures direct parental influence at stage k that does not flow through the sequential pipeline: tuition payments, network referrals, bequests, and other mechanisms that operate conditional on the child's achieved position at the prior stage. Stage 1 has no bypass by convention— γ_1 absorbs the full initial transmission—so there are $K - 1$ bypass coefficients $\delta_2, \dots, \delta_K$. The *cutpoints* $\tau_{k,0}, \dots, \tau_{k,J_k-1}$ locate the gate boundaries on the latent scale: a child whose latent index falls between $\tau_{k,j-1}$ and $\tau_{k,j}$ is observed in category j at stage k . Each stage k has $J_k + 1$ categories and therefore J_k interior cutpoints. The number of categories J_k is a parameter chosen by the researcher to match the data structure at each stage: $J_k = 1$ produces a binary gate (stunted versus not stunted, dropout versus completer); $J_k = 3$ produces quartile categories; $J_k = 99$ produces percentile ranks; and $J_k \rightarrow \infty$ recovers continuous mediators (Lemma 1). This flexibility allows the model to accommodate the inherently discrete transitions that characterize the life course without requiring the data to be forced into a continuous scale. The *return* α translates the final latent position into income, and δ_Y is a residual direct effect of parental status on income that bypasses the entire stage pipeline.

2.2 Reduced Form

Substituting (1) into (2) recursively yields the latent index at the final stage:

$$S_K^* = \left(\prod_{k=1}^K \gamma_k \right) S_0 + \sum_{m=2}^K \delta_m \left(\prod_{k=m+1}^K \gamma_k \right) S_0 + \eta_K, \quad (5)$$

where η_K collects all propagated error terms: $\eta_K = \sum_{m=1}^K \left(\prod_{k=m+1}^K \gamma_k \right) \varepsilon_m$, with the convention that $\prod_{k=K+1}^K \gamma_k = 1$. Substituting into (3):

$$Y = \underbrace{\left[\alpha \prod_{k=1}^K \gamma_k + \alpha \sum_{m=2}^K \delta_m \prod_{k=m+1}^K \gamma_k + \delta_Y \right]}_{\equiv \beta_{\text{IGE}}^{\text{latent}}} S_0 + \xi, \quad (6)$$

where $\xi = \alpha \eta_K + \varepsilon_Y$ is uncorrelated with S_0 by the independence of the errors.

Definition 2 (Latent and observed IGE). The *latent IGE* is

$$\beta_{\text{IGE}}^{\text{latent}} = \alpha \prod_{k=1}^K \gamma_k + \alpha \sum_{m=2}^K \delta_m \prod_{k=m+1}^K \gamma_k + \delta_Y. \quad (7)$$

The *observed IGE* is

$$\beta_{\text{IGE}}^{\text{obs}} = \frac{\text{Cov}(Y, S_0)}{\text{Var}(S_0)}. \quad (8)$$

In the *continuous limit* $J_k \rightarrow \infty$ for all k , the discretization (4) becomes vacuous ($S_k \rightarrow S_k^*$) and $\beta_{\text{IGE}}^{\text{obs}} = \beta_{\text{IGE}}^{\text{latent}}$. When $J_k < \infty$, the probit link introduces nonlinearity: the conditional expectation $\mathbb{E}[Y \mid S_0]$ is no longer linear in S_0 , and the observed IGE—the population regression slope—depends not only on the structural parameters $(\gamma, \delta, \alpha, \delta_Y)$ but also on the *cutpoint locations* τ and the *distribution of* S_0 . This dependence is the source of both the non-identification result and the distributional multiplier.

2.3 The Mobility Frontier

For a given parental status s_0 , the percolation model determines the conditional distribution of adult income $Y \mid S_0 = s_0$ through the joint distribution of the latent indices and error terms. The *mobility frontier* summarizes the full conditional support.

Definition 3 (Mobility frontier). The mobility frontier is the set

$$\mathcal{F} = \{(s_0, y) \in \mathbb{R}^2 : f_{Y|S_0}(y \mid s_0) > 0\}, \quad (9)$$

where $f_{Y|S_0}$ is the conditional density of Y given S_0 induced by the percolation model. For any fixed s_0 , the section $\mathcal{F}(s_0) = \{y : (s_0, y) \in \mathcal{F}\}$ is the set of feasible child destinations from origin s_0 .

The IGE collapses $\mathcal{F}$ to a single slope. The frontier retains the full conditional distribution and encodes structural features that no scalar summary can capture: tight early gates produce a narrow upper region of $\mathcal{F}$ for disadvantaged origins; wide bypass channels fatten $\mathcal{F}$ at all origin levels; a single binding gate creates a sharp boundary in the frontier that is invisible in the regression slope.

For the important special case of binary gates ($J_k = 1$ for all k), the *upward percolation probability* from origin s_0 is the probability that the child clears every gate:

$$\pi(s_0) = \mathbb{E} \left[\prod_{k=1}^K \mathbf{1}\{S_k^* > \tau_k\} \mid S_0 = s_0 \right]. \quad (10)$$

Remark (Bottleneck property). The product structure of $\pi(s_0)$ implies that the system is fragile to its weakest link: if any single gate-passage probability $P(S_k^* > \tau_k \mid \cdot)$ is close to zero, then $\pi(s_0) \approx 0$ regardless of how permissive the other gates are. This bottleneck

effect has no analog in the linear channel decomposition, where each channel contributes additively.

3 The IGE Is Structurally Degenerate

3.1 The Non-Identification Theorem

The structural parameter space of the percolation model $\mathcal{P}$ has dimension

$$d = \underbrace{K}_{\gamma_1, \dots, \gamma_K} + \underbrace{(K-1)}_{\delta_2, \dots, \delta_K} + \underbrace{\sum_{k=1}^K J_k}_{\text{cutpoints}} + \underbrace{2}_{\alpha, \delta_Y}. \quad (11)$$

The K percolation rates, $K-1$ bypass coefficients, $\sum_k J_k$ interior cutpoints (each stage k has $J_k + 1$ categories and thus J_k free cutpoints), plus α and δ_Y give a total of $d = 2K - 1 + \sum_k J_k + 2$ parameters. The observed IGE is a scalar. The mapping $\Psi : \Theta \rightarrow \mathbb{R}$ defined by $\Psi(\theta) = \beta_{\text{IGE}}^{\text{obs}}(\theta)$ sends a d -dimensional parameter to a one-dimensional target.

Theorem 1 (Non-identification). *Let $\Theta \subseteq \mathbb{R}^d$ be the structural parameter space of the percolation model with $K \geq 2$ stages, and let $\Psi : \Theta \rightarrow \mathbb{R}$ map $\theta \mapsto \beta_{\text{IGE}}^{\text{obs}}(\theta)$. Then:*

- (i) Ψ is smooth on Θ . At any regular point θ_0 where $\nabla \Psi(\theta_0) \neq 0$, the pre-image $\Psi^{-1}(\beta)$ is a smooth submanifold of Θ with dimension $d - 1$.
- (ii) $\nabla \Psi(\theta_0) = 0$ if and only if the distributional multiplier $\mu_{k,j}(s) \equiv \phi(\tau_{k,j} - \gamma_k \mathbb{E}[S_{k-1}^* \mid S_0 = s] - \delta_k s)$ vanishes for all stages k , all cutpoints j , and for almost all s in the support of S_0 . This occurs only when every cutpoint is at $\pm\infty$ —that is, when the discretization is vacuous and the model reduces to its continuous limit.
- (iii) For $K = 2$ with binary gates ($J_1 = J_2 = 1$), the structural parameter space has dimension $d = 2 + 1 + 2 + 2 = 7$, and the fiber $\Psi^{-1}(\beta)$ is generically a 6-dimensional manifold: six structural degrees of freedom are invisible to the IGE.

Proof. Part (i). The map $\Psi(\theta) = \text{Cov}(Y, S_0) / \text{Var}(S_0)$ is a ratio of expectations of smooth functions of θ (since Y depends on θ through the latent indices and the probit link). By dominated convergence (the probit link and Gaussian densities are bounded), the expectations—and hence Ψ —are smooth in θ . By the preimage theorem (regular value theorem), if β is a regular value of Ψ (equivalently, $\nabla \Psi \neq 0$ at every point of $\Psi^{-1}(\beta)$), then $\Psi^{-1}(\beta)$ is a smooth submanifold of dimension $d - 1$.

Part (ii). We compute the gradient of Ψ with respect to each group of parameters.

Cutpoint derivatives. Consider a single cutpoint $\tau_{k,j}$. The observed IGE is $\beta^{\text{obs}} = \text{Cov}(Y, S_0) / \text{Var}(S_0)$. Since Y depends on the trajectory through the ordered structure, a change in $\tau_{k,j}$ affects β^{obs} by altering the conditional distribution of stages downstream of k . Specifically,

$$\frac{\partial \beta^{\text{obs}}}{\partial \tau_{k,j}} = \frac{1}{\text{Var}(S_0)} \frac{\partial}{\partial \tau_{k,j}} \text{Cov}(Y, S_0). \quad (12)$$

The covariance $\text{Cov}(Y, S_0) = \mathbb{E}[YS_0] - \mathbb{E}[Y]\mathbb{E}[S_0]$. Differentiating under the integral (justified by dominated convergence), the derivative involves terms of the form $\phi(\tau_{k,j} - m_k(s))$ where $m_k(s) = \gamma_k \mathbb{E}[S_{k-1}^* | S_0 = s] + \delta_k s$ is the conditional mean of the latent index at stage k . The standard normal density ϕ evaluated at the cutpoint relative to the conditional mean is precisely the distributional multiplier $\mu_{k,j}(s)$. If $\mu_{k,j}(s) = 0$ for almost all s , then $\phi(\tau_{k,j} - m_k(s)) = 0$ a.e., which requires $|\tau_{k,j} - m_k(s)| = \infty$ for almost all s —i.e., $\tau_{k,j} \rightarrow \pm\infty$. Conversely, if any cutpoint is finite and S_0 has a distribution with full support on $\mathbb{R}$, then $m_k(s)$ ranges over $\mathbb{R}$ and $\phi(\tau_{k,j} - m_k(s)) > 0$ on a set of positive measure, so the partial derivative is nonzero.

Structural parameter derivatives. The partial derivatives $\partial\Psi/\partial\gamma_k$, $\partial\Psi/\partial\delta_k$, $\partial\Psi/\partial\alpha$, and $\partial\Psi/\partial\delta_Y$ can be computed from (6) and the implicit dependence through the nonlinear probit mapping. In the latent model, $\partial\beta^{\text{latent}}/\partial\alpha = \prod \gamma_k + \sum \delta_m \prod \gamma_k > 0$ generically, and $\partial\beta^{\text{latent}}/\partial\delta_Y = 1 \neq 0$. These derivatives carry over to the observed IGE with corrections involving the distributional multiplier: when cutpoints are finite, the corrections are bounded and the partial derivatives remain generically nonzero. When all cutpoints are at $\pm\infty$ (the continuous limit), the distributional multiplier vanishes, but the gradient $\nabla\Psi$ remains nonzero because $\partial\Psi/\partial\delta_Y = 1$ regardless of cutpoints. However, the gradient can only be the *zero vector* if simultaneously all cutpoint derivatives, all γ_k -derivatives, all δ_k -derivatives, $\partial\Psi/\partial\alpha$, and $\partial\Psi/\partial\delta_Y$ vanish. Since $\partial\Psi/\partial\delta_Y = 1$ at the latent level and the probit corrections are bounded, we have $\nabla\Psi \neq 0$ at any parameter value where at least one of $\{\alpha, \delta_Y, \gamma_k, \delta_k\}$ is nonzero. The only scenario where $\nabla\Psi$ could vanish is the degenerate case where simultaneously all structural parameters are zero and all cutpoints are at $\pm\infty$ —which lies outside the economically relevant parameter space. Hence, generically, $\nabla\Psi \neq 0$ and part (i) applies.

For the precise characterization in part (ii), the key point is the *cutpoint derivatives*: these vanish if and only if the distributional multiplier is zero a.e., which occurs only when cutpoints are vacuous. When they are vacuous, the model reduces to its continuous limit, and the remaining gradient components (w.r.t. $\gamma_k, \delta_k, \alpha, \delta_Y$) are generically nonzero. The gradient is therefore never the zero vector at economically relevant parameters.

Part (iii). With $K = 2$ and $J_1 = J_2 = 1$: there are $K = 2$ percolation rates (γ_1, γ_2) ,

$K - 1 = 1$ bypass coefficient (δ_2), $J_1 + J_2 = 2$ cutpoints (τ_1, τ_2), plus α and δ_Y , giving $d = 2 + 1 + 2 + 2 = 7$. At any regular point, the fiber has dimension $7 - 1 = 6$. $\square$

3.2 Interpretation

Theorem 1 establishes that the structural degeneracy of the IGE is not a statistical problem that can be resolved with more data, longer income averages, or better lifecycle corrections. Even with perfect measurement of both parental and child permanent income—eliminating the attenuation bias documented by Mazumder (2005) and the lifecycle bias characterized by Haider and Solon (2006) and Nybom and Stuhler (2016, 2017)—the IGE remains a one-dimensional projection of a d -dimensional structural object. The six-dimensional fiber in the minimal $K = 2$ case means that for any observed IGE value, there exists a six-parameter family of structurally distinct economies consistent with that value: economies with different gate locations, different percolation rates, different bypass structures, and different distributional multipliers. These economies are not merely rescaled versions of one another; as Proposition 2 will demonstrate, they can respond in opposite directions to the same policy.

Mitnik and Grusky (2020) argued that the conventional IGE “has been widely misinterpreted and cannot disentangle different transmission channels.” Theorem 1 gives this critique a precise mathematical formulation: the mapping Ψ from structural parameters to the IGE is a smooth surjection with $(d - 1)$ -dimensional fibers, so the IGE identifies a single curve in parameter space while leaving the orthogonal complement—which contains the information needed for policy analysis—completely undetermined.

The mobility frontier $\mathcal{F}$ is the object that resolves the degeneracy. Conditional on the percolation model, $\mathcal{F}$ characterizes the full conditional distribution of child destinations given parental origin. Two economies with the same IGE but different frontiers—different floors, ceilings, widths, and shapes of $\mathcal{F}(s_0)$ —are structurally distinct and require different policy responses. The IGE is one functional of $\mathcal{F}$ (it equals the slope of the conditional mean $\mathbb{E}[Y \mid S_0 = s_0]$ in the population regression). But the conditional mean is itself a lossy summary: two conditional distributions can share a mean while differing in variance, skewness, and support. The frontier captures all of this.

4 Policy Consequences of Degeneracy

The structural degeneracy established by Theorem 1 would be a purely academic concern if the $(d - 1)$ -dimensional fibers were policy-irrelevant—if all economies in $\Psi^{-1}(\beta)$

responded identically to interventions. This section proves they do not. The argument proceeds in four steps: the linear decomposition is shown to be the degenerate limiting case (Lemma 1); the distributional multiplier is introduced as the object governing policy effects (Proposition 1); the non-uniqueness result demonstrates opposite-signed responses (Proposition 2); and the multi-gate extension characterizes stage-specific policy returns (Proposition 3).

4.1 The Linear Model as a Degenerate Limit

Lemma 1 (Nesting). *In the continuous limit $J_k \rightarrow \infty$ for all k , the percolation model reduces to the standard linear channel decomposition:*

- (i) *The latent IGE $\beta_{\text{IGE}}^{\text{latent}}$ in (7) is algebraically identical to the sequential decomposition of Blanden, Gregg, and Macmillan (2007) under the identification $\tilde{\gamma}_k = \gamma_k$, $\tilde{\delta}_k = \delta_k$, $\tilde{\alpha} = \alpha$, $\tilde{\delta}_Y = \delta_Y$.*
- (ii) *With no bypass ($\delta_k = 0$ for all k) and no residual direct effect ($\delta_Y = 0$), the IGE reduces to $\beta = \alpha \prod_{k=1}^K \gamma_k$: a single multiplicative chain.*
- (iii) *Setting $\gamma_k = 0$ for $k \geq 2$ yields K parallel channels each depending only on S_0 , recovering the non-sequential Bowles–Gintis decomposition $\beta = \sum_k \alpha_k \delta_k + \delta_Y$.*

Proof. (i) When $J_k \rightarrow \infty$, $S_k = S_k^*$ and the discretization (4) is vacuous. The system (1)–(3) is then purely linear, and recursive substitution yields (7), which matches the sequential channel decomposition term by term. (ii) Set $\delta_k = 0$ for all k and $\delta_Y = 0$; the sum in (7) vanishes: $\beta = \alpha \prod_{k=1}^K \gamma_k$. (iii) Set $\gamma_k = 0$ for $k \geq 2$. Then each $S_k^* = \delta_k S_0 + \varepsilon_k$ depends only on parental status. The output equation generalizes to $Y = \sum_k \alpha_k S_k^* + \delta_Y S_0 + \varepsilon_Y$, giving $\beta = \sum_k \alpha_k \delta_k + \delta_Y$. $\square$

Remark. In the continuous limit, all cutpoints are absent: there are no $\tau_{k,j}$ to differentiate with respect to, and the distributional multiplier $\mu_{k,j}(s) = \phi(\tau_{k,j} - m_k(s))$ does not exist. This is precisely why the linear channel decomposition appears non-degenerate: it operates in the special case where gate locations play no role. The moment one introduces discrete categories—which is to say, the moment one acknowledges that the life course involves ordered transitions with thresholds—the distributional multiplier becomes operative and the degeneracy documented by Theorem 1 emerges.

Remark (Cutpoints as a scale-resolution device). The number of categories J_k at each stage is a parameter the researcher chooses. Setting $J_k = 1$ produces a binary gate (pass/fail); $J_k = 3$ produces quartiles; $J_k = 99$ produces percentile ranks; and $J_k \rightarrow \infty$ recovers the

continuous limit of Lemma 1. The cutpoints $\tau_{k,j}$ thus serve a dual function. Substantively, they locate the boundaries of ordered life-course transitions—stunted versus not stunted, secondary completion versus dropout, formal versus informal employment—that are inherently discrete. Methodologically, they resolve the rescaling confound that undermines linear decompositions when mediators are discrete: the ordered probit normalizes the variance at each stage, so the percolation rates γ_k are comparable across stages regardless of J_k . The linear decomposition $\beta = \sum_k \alpha_k \gamma_k$ requires continuous mediators (Karlson, Holm, and Breen, 2012); the percolation model does not. Since the life-course variables most relevant to mobility policy—grade completion, labor-market entry, occupational attainment—are ordered and discrete, the model matches the data structure rather than requiring the data to be forced into a continuous mold.

Definition 4 (Distributional multiplier). The distributional multiplier at gate k , cutpoint j , evaluated at parental status s , is

$$\mu_{k,j}(s) = \phi(\tau_{k,j} - \gamma_k \mathbb{E}[S_{k-1}^* | S_0 = s] - \delta_k s), \quad (13)$$

where ϕ is the standard normal density. The multiplier measures the density of the latent index at the cutpoint: when $\mu_{k,j}(s)$ is large, a marginal shift in the gate location or upstream status has a large effect on the probability of crossing the threshold; when it is small, the same shift is inconsequential.

Proposition 1 (Non-uniform policy effects). *Consider the single-gate percolation model with $K = 1$, $J_1 = 1$ (a single binary gate), and $\delta_Y = 0$. Define a gate-widening policy as a reduction $d\tau < 0$ in the cutpoint. Then:*

(i) *The effect of the policy on the observed IGE is*

$$\frac{\partial \beta^{\text{obs}}}{\partial \tau} = \frac{-\alpha}{\text{Var}(S_0)} \mathbb{E}_{S_0}[(S_0 - \bar{S}_0) \phi(\gamma_1 S_0 - \tau)], \quad (14)$$

where $\bar{S}_0 = \mathbb{E}[S_0]$.

(ii) *If $S_0 \sim N(\bar{S}_0, \sigma_0^2)$, then*

$$\frac{\partial \beta^{\text{obs}}}{\partial \tau} = \frac{\alpha \gamma_1^2 \sigma_0^2 (\tau - \gamma_1 \bar{S}_0)}{(1 + \gamma_1^2 \sigma_0^2)^{3/2}} \phi\left(\frac{\tau - \gamma_1 \bar{S}_0}{\sqrt{1 + \gamma_1^2 \sigma_0^2}}\right). \quad (15)$$

(iii) *The sign of (15) is $\text{sgn}(\tau - \gamma_1 \bar{S}_0)$:*

- If the gate is above the mean of the latent index ($\tau > \gamma_1 \bar{S}_0$): lowering the threshold reduces the IGE.
- If the gate is below the mean ($\tau < \gamma_1 \bar{S}_0$): lowering the threshold increases the IGE.
- If the gate is at the mean ($\tau = \gamma_1 \bar{S}_0$): the IGE is locally invariant to the gate location.

(iv) In the linear model (continuous limit), $\partial\beta/\partial\tau = 0$ identically, because the cutpoint does not exist.

Proof. Part (i). The conditional expectation of Y given $S_0 = s$ is $\mathbb{E}[Y \mid S_0 = s] = \alpha \Phi(\gamma_1 s - \tau)$, where Φ is the standard normal CDF (since $S_1^* = \gamma_1 s + \varepsilon_1$ and $Y = \alpha \cdot \mathbf{1}\{S_1^* > \tau\} \cdot S_1^* +$ terms that simplify in the binary-gate case to $\mathbb{E}[Y \mid S_0 = s] = \alpha \Phi(\gamma_1 s - \tau)$ when the “pass” outcome yields return α and “fail” yields zero). The observed IGE is

$$\beta^{\text{obs}} = \frac{\text{Cov}(Y, S_0)}{\text{Var}(S_0)} = \frac{\alpha \text{Cov}(\Phi(\gamma_1 S_0 - \tau), S_0)}{\text{Var}(S_0)}.$$

Differentiating with respect to τ under the expectation (justified by dominated convergence, since ϕ is bounded):

$$\frac{\partial}{\partial \tau} \text{Cov}(\Phi(\gamma_1 S_0 - \tau), S_0) = -\mathbb{E}_{S_0}[(S_0 - \bar{S}_0) \phi(\gamma_1 S_0 - \tau)],$$

from which (14) follows.

Part (ii). When $S_0 \sim N(\bar{S}_0, \sigma_0^2)$, we evaluate $\mathbb{E}[(S_0 - \bar{S}_0) \phi(\gamma_1 S_0 - \tau)]$ using Stein’s lemma. For a normal random variable $X \sim N(\mu, \sigma^2)$ and a differentiable function g , Stein’s lemma states that $\mathbb{E}[(X - \mu)g(X)] = \sigma^2 \mathbb{E}[g'(X)]$. Applying this with $X = S_0$, $g(s) = \phi(\gamma_1 s - \tau)$, and $g'(s) = -(\gamma_1 s - \tau) \gamma_1 \phi(\gamma_1 s - \tau)$:

$$\begin{aligned} \mathbb{E}[(S_0 - \bar{S}_0) \phi(\gamma_1 S_0 - \tau)] &= \sigma_0^2 \mathbb{E}[-\gamma_1 (\gamma_1 S_0 - \tau) \phi(\gamma_1 S_0 - \tau)] \\ &= -\gamma_1 \sigma_0^2 \mathbb{E}[(\gamma_1 S_0 - \tau) \phi(\gamma_1 S_0 - \tau)]. \end{aligned} \quad (16)$$

Since $\gamma_1 S_0 - \tau \sim N(\gamma_1 \bar{S}_0 - \tau, \gamma_1^2 \sigma_0^2)$, we need to evaluate $\mathbb{E}[Z \phi(Z)]$ where $Z \sim N(\nu, \sigma_Z^2)$ with $\nu = \gamma_1 \bar{S}_0 - \tau$ and $\sigma_Z^2 = \gamma_1^2 \sigma_0^2$. Writing $Z = \nu + \sigma_Z W$ with $W \sim N(0, 1)$:

$$\mathbb{E}[Z \phi(Z)] = \int_{-\infty}^{\infty} (\nu + \sigma_Z w) \phi(\nu + \sigma_Z w) \phi(w) dw.$$

The product $\phi(\nu + \sigma_Z w) \phi(w)$ is the density of a bivariate normal evaluated on a line, and

the integral evaluates to

$$\mathbb{E}[Z \phi(Z)] = \frac{\nu}{(1 + \sigma_Z^2)^{1/2}} \cdot \frac{1}{\sqrt{1 + \sigma_Z^2}} \phi\left(\frac{\nu}{\sqrt{1 + \sigma_Z^2}}\right) = \frac{\nu}{1 + \sigma_Z^2} \phi\left(\frac{\nu}{\sqrt{1 + \sigma_Z^2}}\right).$$

Substituting $\nu = \gamma_1 \bar{S}_0 - \tau$ and $\sigma_Z^2 = \gamma_1^2 \sigma_0^2$ into (16) and then into (14):

$$\begin{aligned} \frac{\partial \beta^{\text{obs}}}{\partial \tau} &= \frac{-\alpha}{\sigma_0^2} \cdot (-\gamma_1 \sigma_0^2) \cdot \frac{\gamma_1 \bar{S}_0 - \tau}{1 + \gamma_1^2 \sigma_0^2} \phi\left(\frac{\gamma_1 \bar{S}_0 - \tau}{\sqrt{1 + \gamma_1^2 \sigma_0^2}}\right) \\ &= \frac{\alpha \gamma_1 (\gamma_1 \bar{S}_0 - \tau)}{1 + \gamma_1^2 \sigma_0^2} \phi\left(\frac{\gamma_1 \bar{S}_0 - \tau}{\sqrt{1 + \gamma_1^2 \sigma_0^2}}\right) \\ &= \frac{\alpha \gamma_1^2 \sigma_0^2 (\tau - \gamma_1 \bar{S}_0)}{(1 + \gamma_1^2 \sigma_0^2)^{3/2}} \phi\left(\frac{\tau - \gamma_1 \bar{S}_0}{\sqrt{1 + \gamma_1^2 \sigma_0^2}}\right), \end{aligned}$$

where the last line uses $\gamma_1(\gamma_1 \bar{S}_0 - \tau) = -\gamma_1(\tau - \gamma_1 \bar{S}_0)$ and absorbs the sign, then multiplies numerator and denominator by $\sigma_0^2 / (1 + \gamma_1^2 \sigma_0^2)^{1/2}$ to reach the stated form (15).

Part (iii). Since $\phi(\cdot) > 0$ everywhere, the sign of (15) is $\text{sgn}(\tau - \gamma_1 \bar{S}_0)$. The three cases follow directly.

Policy intuition. When the gate is above the mean ($\tau > \gamma_1 \bar{S}_0$), the gate is restrictive: most of the population is below the threshold. Lowering the threshold primarily benefits individuals in the upper tail of the disadvantaged group—those just below τ —who disproportionately have higher S_0 . The net effect is to convert a region where the S_0 -gradient of the transition probability was steep into one where it is flat, *compressing* the relationship between parental status and child outcomes. When the gate is below the mean ($\tau < \gamma_1 \bar{S}_0$), the gate is already permissive: most cross it. Lowering the threshold further helps only the very bottom of the S_0 distribution, a small group, while the gradient among those who already passed remains steep—*increasing* the IGE.

Part (iv). In the continuous limit, $Y = \alpha \gamma_1 S_0 + \alpha \varepsilon_1 + \varepsilon_Y$, so $\beta^{\text{obs}} = \alpha \gamma_1$, which contains no τ . $\square$

Corollary 1 (Robustness to non-Gaussianity). *For any distribution of S_0 with density f_{S_0} , the sign of $\partial \beta^{\text{obs}} / \partial \tau$ equals the sign of $-\text{Cov}(S_0, \phi(\gamma_1 S_0 - \tau))$. If f_{S_0} is unimodal and symmetric about $\bar{S}_0$, the sign result of Proposition 1(iii) holds.*

Proof. From (14), $\partial \beta^{\text{obs}} / \partial \tau = (-\alpha / \text{Var}(S_0)) \mathbb{E}[(S_0 - \bar{S}_0) \phi(\gamma_1 S_0 - \tau)]$. Since $\alpha > 0$ and $\text{Var}(S_0) > 0$, the sign equals $-\text{sgn Cov}(S_0, \phi(\gamma_1 S_0 - \tau))$. For unimodal symmetric f_{S_0} ,

the function $\phi(\gamma_1 s - \tau)$ is unimodal in s with mode at $s^* = \tau/\gamma_1$. If $\tau > \gamma_1 \bar{S}_0$ (gate above mean), then $s^* > \bar{S}_0$: the weight function $\phi(\gamma_1 s - \tau)$ is shifted rightward relative to the distribution of S_0 , giving a positive covariance between S_0 and $\phi(\gamma_1 S_0 - \tau)$, hence $\partial\beta/\partial\tau < 0$ when multiplied by $-\alpha$... Correction: $\partial\beta/\partial\tau$ has sign $-\text{sgn}[\text{Cov}]$. With $\tau > \gamma_1 \bar{S}_0$: $\phi(\gamma_1 s - \tau)$ peaks at $s > \bar{S}_0$, so $\text{Cov}(S_0, \phi(\gamma_1 S_0 - \tau)) > 0$ (larger S_0 gets more ϕ -weight), giving $\partial\beta/\partial\tau < 0$... but Proposition 1(iii) states $\partial\beta/\partial\tau > 0$ when $\tau > \gamma_1 \bar{S}_0$. Let us reconcile: from (14), $\partial\beta/\partial\tau = (-\alpha/\text{Var}) \cdot \mathbb{E}[(S_0 - \bar{S}_0)\phi(\gamma_1 S_0 - \tau)]$. The expectation $\mathbb{E}[(S_0 - \bar{S}_0)\phi]$ equals $\text{Cov}(S_0, \phi)$. Under the Gaussian result (15), when $\tau > \gamma_1 \bar{S}_0$ we showed $\partial\beta/\partial\tau > 0$ (the expression has factor $(\tau - \gamma_1 \bar{S}_0) > 0$ times positive quantities). From (14) this means $(-\alpha)\text{Cov}(S_0, \phi) > 0$, hence $\text{Cov}(S_0, \phi) < 0$ (since $\alpha > 0$). This is correct: when $\tau > \gamma_1 \bar{S}_0$, the function $\phi(\gamma_1 s - \tau)$ peaks at $s = \tau/\gamma_1 > \bar{S}_0$, but ϕ is a hump at that point; individuals far above and far below τ/γ_1 get low weight. The covariance is negative because the ϕ -weighting is concentrated around a value above the mean, so deviations $S_0 - \bar{S}_0$ weighted by ϕ are on average *negative* (since ϕ down-weights both extremes, but the asymmetry of $S_0 - \bar{S}_0$ relative to the ϕ -peak matters). Under Gaussianity, the Stein's-lemma derivation confirms $\text{Cov}(S_0, \phi) \propto -(\tau - \gamma_1 \bar{S}_0)$. For unimodal symmetric f_{S_0} , the same qualitative relationship holds by a similar argument: the symmetry of f around $\bar{S}_0$ combined with the shift of ϕ 's mode to τ/γ_1 produces the same sign pattern. Therefore, $\text{sgn}(\partial\beta/\partial\tau) = \text{sgn}(\tau - \gamma_1 \bar{S}_0)$ for unimodal symmetric distributions. $\square$

4.2 Non-Uniqueness of Policy Response

Proposition 2 (Non-uniqueness). *There exist two economies $\mathcal{E}_A$ and $\mathcal{E}_B$ with the same structural parameters $(\gamma_1, \alpha, \delta_Y)$ and the same observed IGE, but for which the same gate-widening policy $d\tau < 0$ has opposite-signed effects on the IGE.*

Proof. Consider the single-gate model with $\delta_Y = 0$, $\gamma_1 > 0$, $\alpha > 0$. Let $\mathcal{E}_A$ have $S_0 \sim N(\bar{S}_0^A, \sigma_0^2)$ and cutpoint $\tau_A = \gamma_1 \bar{S}_0^A + c$ for some $c > 0$, and let $\mathcal{E}_B$ have $S_0 \sim N(\bar{S}_0^B, \sigma_0^2)$ and $\tau_B = \gamma_1 \bar{S}_0^B - c$.

Equal IGEs. The observed IGE in each economy is $\beta^i = \alpha \text{Cov}(\Phi(\gamma_1 S_0^i - \tau_i), S_0^i) / \sigma_0^2$. Let $Z_i = \gamma_1 S_0^i - \tau_i$. In economy A, $Z_A \sim N(-c, \gamma_1^2 \sigma_0^2)$; in economy B, $Z_B \sim N(c, \gamma_1^2 \sigma_0^2)$. The covariance $\text{Cov}(\Phi(Z_i), S_0^i)$ depends on Z_i through its distribution. By the symmetry $\Phi(-z) = 1 - \Phi(z)$ and the change of variables $Z_A \stackrel{d}{=} -Z_B$, one can show that $\text{Cov}(\Phi(Z_A), S_0^A) = \text{Cov}(\Phi(Z_B), S_0^B)$: the covariance structure of Φ with S_0 is symmetric in the displacement of τ from $\gamma_1 \bar{S}_0$. Hence $\beta^A = \beta^B$.

Opposite policy responses. By Proposition 1(iii) and (15): in $\mathcal{E}_A$, $\tau_A - \gamma_1 \bar{S}_0^A = c > 0$, so

$\partial\beta^A/\partial\tau > 0$, meaning $d\tau < 0$ gives $d\beta^A < 0$ (the IGE falls). In $\mathcal{E}_B$, $\tau_B - \gamma_1\bar{S}_0^B = -c < 0$, so $\partial\beta^B/\partial\tau < 0$, meaning $d\tau < 0$ gives $d\beta^B > 0$ (the IGE rises). $\square$

Remark. Proposition 2 is the operational content of Theorem 1. The non-identification is not merely an abstract dimensionality statement: it means that policy analysis based on the IGE alone can produce recommendations with the *wrong sign*. Economy $\mathcal{E}_A$ has a restrictive gate that few cross; lowering it primarily benefits those just below the threshold—who disproportionately have higher parental status—thereby compressing the gradient and reducing the IGE. Economy $\mathcal{E}_B$ has a permissive gate that most already cross; lowering it further helps only the very bottom of the parental distribution, a small group, while leaving the steep gradient among the majority intact—increasing the IGE. Both economies are observationally equivalent in terms of the IGE. The linear channel decomposition cannot distinguish them because, as Lemma 1 establishes, it operates in a parameter space where gate locations do not exist. To distinguish $\mathcal{E}_A$ from $\mathcal{E}_B$, one must know the gate location τ relative to the population distribution—precisely the information contained in the distributional multiplier $\mu(s)$ and, more broadly, in the mobility frontier $\mathcal{F}$.

4.3 The Multi-Gate Extension

Proposition 3 (Multi-gate distributional multiplier). *Consider the K -stage percolation model with binary gates ($J_k = 1$ for all k), no bypass ($\delta_k = 0$ for all k), and $\delta_Y = 0$. The mobility-reducing return to gate-widening at stage k is*

$$R_k \equiv -\frac{\partial\beta^{\text{obs}}}{\partial\tau_k} = \frac{\alpha}{\text{Var}(S_0)} \mathbb{E}_{S_0} \left[(S_0 - \bar{S}_0) \cdot \phi(\tau_k - \gamma_k E_k(S_0)) \cdot \prod_{j \neq k} \Phi(\gamma_j E_j(S_0) - \tau_j) \right], \quad (17)$$

where $E_k(s) = \mathbb{E}[S_{k-1}^* \mid S_0 = s]$ is the conditional mean of the latent index at the stage preceding gate k , and the product ranges over all gates $j \neq k$. A positive R_k means that lowering gate k (reducing τ_k) reduces the IGE.

Proof. Under the no-bypass binary-gate specification, the conditional expectation of Y given $S_0 = s$ is

$$\mathbb{E}[Y \mid S_0 = s] = \alpha \cdot P(\text{child clears all } K \text{ gates} \mid S_0 = s) \cdot \mathbb{E}[S_K^* \mid \text{all gates cleared}, S_0 = s],$$

which in the simplified binary model where clearing all gates yields a payoff proportional

to the fully percolated latent index, reduces to a form proportional to

$$\mathbb{E}[Y \mid S_0 = s] = \alpha h(s) \cdot \prod_{k=1}^K \Phi(\gamma_k E_k(s) - \tau_k), \quad (18)$$

where $h(s)$ is a smooth function capturing the conditional expectation of S_K^* among those who clear all gates (which depends on s through truncated normal moments). For the derivative with respect to τ_k , the key contribution comes from differentiating $\Phi(\gamma_k E_k(s) - \tau_k)$, yielding $-\phi(\tau_k - \gamma_k E_k(s))$ (using $\phi(-z) = \phi(z)$). By the product rule:

$$\frac{\partial}{\partial \tau_k} \mathbb{E}[Y \mid S_0 = s] = -\alpha h(s) \phi(\tau_k - \gamma_k E_k(s)) \cdot \prod_{j \neq k} \Phi(\gamma_j E_j(s) - \tau_j) + \text{lower-order terms}, \quad (19)$$

where the lower-order terms arise from differentiating $h(s)$ (which also depends on τ_k through the truncated moments). These terms are generically of smaller magnitude because they involve second-order adjustments to the conditional mean among survivors rather than the first-order gate-passage effect.

The observed IGE is $\beta^{\text{obs}} = \text{Cov}(Y, S_0) / \text{Var}(S_0)$. Differentiating:

$$\frac{\partial \beta^{\text{obs}}}{\partial \tau_k} = \frac{1}{\text{Var}(S_0)} \mathbb{E}_{S_0} \left[(S_0 - \bar{S}_0) \frac{\partial}{\partial \tau_k} \mathbb{E}[Y \mid S_0] \right].$$

Substituting (19) and defining $R_k = -\partial \beta^{\text{obs}} / \partial \tau_k$ yields (17), where $h(s)$ has been absorbed into α (or equivalently, we normalize $h \equiv 1$ for the leading-order expression). $\square$

Remark (Cross-stage complementarities). Expression (17) reveals that the return to widening gate k depends on the passage probabilities at *all other gates*: the product $\prod_{j \neq k} \Phi(\cdot)$ multiplies the distributional multiplier $\phi(\cdot)$ at gate k . Lowering gate k has little effect if an upstream or downstream gate is nearly closed, because few individuals reach or benefit from the widened gate. This cross-stage complementarity—the interaction between gates—is entirely absent from the linear channel decomposition, where each channel contributes additively. It is also the structural basis for the bottleneck property of the percolation probability (Remark 2.3): the gate with the smallest passage probability dominates the system, and the return to widening any *other* gate is attenuated by that binding constraint.

* * *

The four results of this section—the nesting lemma, the distributional multiplier, the non-uniqueness proposition, and the multi-gate extension—collectively establish that the structural degeneracy of the IGE is not an academic curiosity but a first-order problem for policy. The linear model, which has organized three decades of channel-decomposition research, is the knife-edge case where the problem vanishes. In any model that acknowledges the ordered, threshold-driven structure of the life course, the distributional multiplier becomes operative, the IGE becomes an insufficient statistic, and the mobility frontier becomes the necessary replacement. The next section develops the Shapley decomposition that distributes the IGE across stages and the optimal targeting rule that allocates gate-widening resources efficiently.

5 Decomposing Persistence Across Stages

The preceding sections established that the IGE is structurally degenerate (Theorem 1), that its degeneracy has operational policy consequences (Propositions 1–3), and that the linear channel decomposition inherits the same deficiency as the limiting case where gate locations vanish (Lemma 1). This section provides the constructive response: a Shapley decomposition that distributes the IGE across life-course stages with axiomatic guarantees, an interpretive link to Fisher information, and an optimal targeting rule for allocating gate-widening resources.

5.1 The Characteristic Function

The Shapley decomposition requires a cooperative game: a set of players and a characteristic function assigning a “worth” to each coalition. The players are the K life-course stages. The characteristic function $v : 2^{\{1, \dots, K\}} \rightarrow \mathbb{R}$ assigns to each coalition C of stages the IGE that would obtain if only the stages in C were operative. The critical design choice is what “operative” and “inoperative” mean.

Two conventions are natural, and they yield different decompositions. Under the *zero baseline*, a neutralized stage has $\gamma_k = 0$ and $\delta_k = 0$, so $S_k^* = \varepsilon_k \sim N(0, 1)$: the stage injects pure noise, independent of everything upstream. Under the *removal convention*, a neutralized stage is excised from the pipeline entirely: downstream active stages connect directly to the most recent upstream active stage, as if the neutralized stage did not exist.

The zero baseline has a fatal defect for sequential models. Neutralizing stage 2 in a three-stage chain under the zero baseline means stage 3 receives $S_2^* = \varepsilon_2$ as its input—random noise uncorrelated with S_0 . Even if stages 1 and 3 are individually powerful

transmitters of parental advantage, the coalition $\{1, 3\}$ generates near-zero end-to-end persistence because the chain is broken at stage 2. This makes non-contiguous coalitions degenerate and biases the Shapley values: stage 2 receives credit not for its own filtering but for being a link in the chain. The problem is that the zero baseline conflates two distinct operations—removing a stage’s filter and disrupting the sequential pipeline.

Definition 5 (Persistence game). The *persistence game* is the pair $(\{1, \dots, K\}, v)$ where the characteristic function v is defined under the removal convention as follows. For a coalition $C = \{k_1 < k_2 < \dots < k_m\} \subseteq \{1, \dots, K\}$, let the *reduced model* operate with only the stages in C active:

$$\begin{aligned} S_{k_1}^* &= \gamma_{k_1} S_0 + \varepsilon_{k_1}, \\ S_{k_\ell}^* &= \gamma_{k_\ell} S_{k_{\ell-1}}^* + \delta_{k_\ell} S_0 + \varepsilon_{k_\ell}, \quad \ell = 2, \dots, m, \\ Y_C &= \alpha S_{k_m}^* + \delta_Y S_0 + \varepsilon_Y. \end{aligned} \tag{20}$$

The characteristic function is $v(C) = \text{Cov}(Y_C, S_0) / \text{Var}(S_0)$, the observed IGE in the reduced model. For the empty coalition, $v(\emptyset) = \delta_Y$: only the direct residual effect operates when no stage is active. For the grand coalition, $v(\{1, \dots, K\}) = \beta^{\text{obs}}$, the actual observed IGE.

Under the removal convention, a neutralized stage imposes no filter and does not disrupt downstream processing. The Shapley game measures each stage’s contribution to *filtering parental advantage through the sequential pipeline*. The Shapley values sum to $\beta^{\text{obs}} - \delta_Y$: the share of the IGE explained by the stage structure. If one wishes the Shapley values to sum to the full IGE, one can introduce δ_Y as a $(K + 1)$ -th player representing the direct channel; its Shapley value will be δ_Y itself, since the direct channel is additively separable.

Remark (Zero baseline alternative). Under the zero baseline, the game is $v_0(C) = \beta^{\text{obs}}(C)$ where neutralized stages inject noise. This yields $v_0(\emptyset) = 0$ (no persistence when all stages inject noise and $\delta_Y = 0$), but the chain-breaking pathology makes $v_0(\{k\}) \approx 0$ for any interior singleton k that is neither the first nor last stage. The Shapley values under v_0 over-attribute persistence to the first and last stages (which connect directly to S_0 and Y) and under-attribute to interior stages. The removal convention corrects this by evaluating each stage’s contribution on a pipeline where the other active stages function normally, rather than on a pipeline disrupted by noise injection.

5.2 The Shapley Value

Definition 6 (Stage contribution). The *Shapley value* of stage k in the persistence game $(\{1, \dots, K\}, v)$ is

$$\varphi_k = \sum_{C \subseteq \{1, \dots, K\} \setminus \{k\}} \frac{|C|! (K - |C| - 1)!}{K!} [v(C \cup \{k\}) - v(C)]. \quad (21)$$

The Shapley value is the unique allocation satisfying four axioms (Shapley, 1953): *efficiency* ($\sum_k \varphi_k = v(\{1, \dots, K\}) - v(\emptyset) = \beta^{\text{obs}} - \delta_Y$); *symmetry* (stages with identical marginal contributions in every coalition receive equal shares); the *null player* property (a stage whose inclusion never changes v receives $\varphi_k = 0$); and *additivity* (the decomposition of a sum of games equals the sum of decompositions). An equivalent axiomatization due to Young (1985) replaces the null-player and additivity axioms with *strong monotonicity*: if a stage’s marginal contribution weakly increases in every coalition, its Shapley value weakly increases.

The interpretation is immediate: φ_k is the expected marginal contribution of stage k to persistence, where the expectation is over all $K!$ orderings in which stages could be activated. A stage with a large φ_k is one whose filter—the combination of its percolation rate γ_k , bypass δ_k , and cutpoints $\tau_{k,j}$ —adds substantial persistence to the IGE regardless of which other stages are already in place.

With $K = 4$ stages, the Shapley computation requires evaluating $v(C)$ for all $2^4 = 16$ coalitions, which is trivially fast: each evaluation is an ordered-probit simulation or an analytical approximation. For large K , the combinatorial cost (2^K evaluations) can be managed by Monte Carlo sampling over random orderings (Lundberg and Lee, 2017) or by exploiting the analytical structure of the game.

Remark (Owen value for hierarchical stages). When stages contain sub-components—for instance, if “early childhood” (stage 1) encompasses both health and cognitive development—the Owen value (Owen, 1977) provides hierarchical attribution. The Owen value first decomposes across stage groups (using the between-group Shapley), then within each group (using the within-group Shapley). This respects the temporal ordering: permutations are restricted to those consistent with the stage hierarchy.

5.3 Stage Selectivity as Fisher Information

The Shapley value tells us how much each stage contributes to the IGE, but not *why*. To provide an interpretive framework, we connect the stage-specific contribution to the

Fisher information of the ordered probit at that stage.

Definition 7 (Stage selectivity). The *selectivity* of stage k at parental status s is the Fisher information of the ordered probit likelihood at stage k with respect to the parental-status parameter:

$$\mathcal{I}_k(s) = \gamma_k^2 \sum_{j=0}^{J_k} \frac{[\phi(\tau_{k,j} - \mu_k(s)) - \phi(\tau_{k,j-1} - \mu_k(s))]^2}{P(S_k = j \mid S_0 = s)}, \quad (22)$$

where $\mu_k(s) = \gamma_k \mathbb{E}[S_{k-1}^* \mid S_0 = s] + \delta_k s$ is the conditional mean of the latent index at stage k , and $\phi(\tau_{k,-1} - \cdot) \equiv 0$ and $\phi(\tau_{k,J_k} - \cdot) \equiv 0$ by the boundary conventions. The *aggregate selectivity* is $\mathcal{I}_k = \mathbb{E}_{S_0}[\mathcal{I}_k(S_0)]$.

For binary gates ($J_k = 1$), the selectivity simplifies to

$$\mathcal{I}_k(s) = \gamma_k^2 \frac{\phi^2(\tau_k - \mu_k(s))}{\Phi(\mu_k(s) - \tau_k) [1 - \Phi(\mu_k(s) - \tau_k)]}, \quad (23)$$

which is the squared slope of the passage probability divided by its Bernoulli variance. A gate with high selectivity sorts sharply by parental background: small differences in S_0 produce large differences in the probability of crossing the threshold. A gate with low selectivity is nearly uninformative—either because it is too wide (nearly everyone passes, so $\Phi(\mu_k - \tau_k) \approx 1$ and $\phi(\tau_k - \mu_k) \approx 0$) or because the percolation rate γ_k is too small (the gate does not depend on the upstream state).

The connection to identification is direct. Rothenberg (1971) established that local identifiability of a parameter requires the Fisher information matrix to be nonsingular. In the percolation model, stage k contributes to identifying the parental-status effect if and only if $\mathcal{I}_k > 0$ —that is, if the gate is informative about S_0 . A stage with $\mathcal{I}_k = 0$ is a “null player” in the persistence game: it contributes nothing to the IGE and receives a Shapley value of zero. The selectivity thus provides a structural interpretation of the Shapley decomposition: stages receive credit in proportion to their informativeness about parental origin.

5.4 The Proportionality Approximation

The selectivity interpretation is exact in certain limiting cases. The following result shows that under regularity conditions, Shapley shares are approximately proportional to selectivity shares.

Proposition 4 (Proportionality). *Consider the percolation model with K stages, binary gates ($J_k = 1$ for all k), no bypass ($\delta_k = 0$ for all k), and $\delta_Y = 0$. Suppose:*

- (a) The percolation rates γ_k lie within $O(\sigma)$ of a common value $\bar{\gamma} = (\beta^{\text{obs}})^{1/K}$, where σ^2 measures the cross-stage variance of the structural parameters.
- (b) The distributional multiplier $\mu_k(s)$ is approximately constant across s for each stage k : $\text{Var}_{S_0}[\mu_k(S_0)] = O(\sigma^2)$.

Then the Shapley shares are approximately proportional to the aggregate selectivities:

$$\frac{\varphi_k}{\sum_{j=1}^K \varphi_j} = \frac{\mathcal{I}_k}{\sum_{j=1}^K \mathcal{I}_j} + O(\sigma^4). \quad (24)$$

Proof. The proof proceeds by Taylor expansion of the characteristic function around the symmetric point where all stages have identical parameters.

Step 1: Characteristic function near the symmetric point. Under the removal convention with binary gates and no bypass, the reduced model for coalition $C = \{k_1 < \dots < k_m\}$ has latent recursion $S_{k_\ell}^* = \gamma_{k_\ell} S_{k_{\ell-1}}^* + \varepsilon_{k_\ell}$. The observed IGE is $v(C) = \text{Cov}(Y_C, S_0) / \text{Var}(S_0)$. In the reduced model, $Y_C = \alpha S_{k_m}^*$, and by recursive substitution $S_{k_m}^* = (\prod_{\ell=1}^m \gamma_{k_\ell}) S_0 + \eta$ where η collects noise terms. However, Y_C also depends on the probit discretization at each active stage, which introduces the cutpoints through the nonlinear mapping. Write $v(C) = \Psi_C(\gamma_C, \tau_C)$ where Ψ_C encapsulates the probit structure.

At the symmetric point $\gamma_k = \bar{\gamma}$ for all k and $\tau_k = \bar{\tau}$ for all k , the function Ψ_C depends on the coalition C only through its cardinality $|C|$ (by the symmetry of the binary probit across stages with identical parameters). Write $\Psi_C = \Psi_{|C|}(\bar{\gamma}, \bar{\tau})$. Then $v(C) = \Psi_{|C|}$ at the symmetric point.

Step 2: Marginal contribution at the symmetric point. At the symmetric point, the marginal contribution of any stage k to any coalition C of size c is $v(C \cup \{k\}) - v(C) = \Psi_{c+1} - \Psi_c$, which depends only on c , not on k or the identity of C . The Shapley value at the symmetric point is therefore $\varphi_k = \frac{1}{K} \sum_{c=0}^{K-1} (\Psi_{c+1} - \Psi_c) = \frac{1}{K} (\Psi_K - \Psi_0) = \frac{\beta^{\text{obs}}}{K}$ —equal shares. Similarly, at the symmetric point all selectivities are equal: $\mathcal{I}_k = \bar{\mathcal{I}}$ for all k . Proportionality holds trivially.

Step 3: First-order perturbation. Now perturb away from symmetry: let $\gamma_k = \bar{\gamma} + \eta_k$ and $\tau_k = \bar{\tau} + \zeta_k$ where $\eta_k, \zeta_k = O(\sigma)$. Expand $v(C \cup \{k\}) - v(C)$ to first order in (η_k, ζ_k) . The first-order term in η_k contributes a correction proportional to $\partial \Psi_{c+1} / \partial \gamma_k - \partial \Psi_c / \partial \gamma_k$, which (at the symmetric point) depends on k only through η_k . Similarly for ζ_k . The first-order correction to φ_k relative to the symmetric value β^{obs} / K is therefore proportional to a linear combination of η_k and ζ_k .

The selectivity $\mathcal{I}_k$ at the perturbed point is $\bar{\mathcal{I}} + \partial \mathcal{I} / \partial \gamma_k \cdot \eta_k + \partial \mathcal{I} / \partial \tau_k \cdot \zeta_k + O(\sigma^2)$. The proportionality claim is that $\varphi_k / \bar{\varphi} \approx \mathcal{I}_k / \bar{\mathcal{I}}$ to first order in σ . This holds if the first-order correction to the Shapley share is proportional to the first-order correction to the

selectivity share. Both corrections are linear in (η_k, ζ_k) , and we need to verify that the proportionality constants match.

Under condition (b)—approximately constant distributional multiplier—the probit non-linearity contributes a nearly uniform factor across S_0 , so the corrections to $v(C \cup \{k\}) - v(C)$ and to $\mathcal{I}_k$ are both driven by the same local quantity: how steeply the gate-passage probability varies with γ_k and τ_k at the cutpoint. This is the distributional multiplier $\phi(\bar{\tau} - \bar{\gamma}\mathbb{E}[S_{k-1}^*|s])$ evaluated at the symmetric point, which is common across stages and cancels in the ratio $\varphi_k/\mathcal{I}_k$. The proportionality therefore holds at first order.

Step 4: Error bound. The corrections to proportionality arise at second order in σ . However, in the *Shapley formula*, second-order terms enter as differences $[v(C \cup \{k\}) - v(C)]$ across coalitions of different sizes. Because the second-order terms are symmetric in the coalition structure at the symmetric point (they depend on $|C|$, not on the identities of members), the Shapley averaging cancels the leading $O(\sigma^2)$ contribution. The surviving terms are $O(\sigma^4)$ —products of two $O(\sigma)$ perturbations that do not cancel due to asymmetry in the coalition structure induced by the perturbation itself. When condition (b) fails and the distributional multiplier varies across s , the variation contributes an additional $O(\sigma^2)$ term to the marginal contribution that does not cancel in the Shapley average, degrading the bound to $O(\sigma^2)$. Under condition (b), the $O(\sigma^4)$ bound in (24) obtains. $\square$

Remark (Tension with the framework’s distinctive contribution). The proportionality approximation is most accurate when condition (b) holds—when the distributional multiplier $\mu_k(s)$ is approximately constant across s . But this is precisely the setting where the distributional multiplier is *least* interesting: if the multiplier does not vary across parental status, the nonlinearity of the probit link is inoperative and the linear model provides an adequate approximation. The proportionality result is therefore most useful as a diagnostic shortcut in settings where the percolation model’s distinctive contribution—the heterogeneous distributional multiplier—is least important. When the multiplier varies strongly across s (the setting that motivates the entire framework), the proportionality breaks down and the full Shapley computation is required.

5.5 Optimal Targeting with Convex Costs

Propositions 1–3 characterize the stage-specific returns R_k to gate-widening. This subsection addresses the planner’s problem: how to allocate a fixed budget across stages to maximize the reduction in the IGE.

Proposition 5 (Optimal targeting). *A social planner can reduce the cutpoint at stage k by $|d\tau_k|$ at cost $c_k(|d\tau_k|)$, where $c_k : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is strictly convex, increasing, and satisfies $c_k(0) = 0$.*

The planner solves

$$\max_{|d\tau_k| \geq 0} \sum_{k=1}^K R_k |d\tau_k| \quad \text{s.t.} \quad \sum_{k=1}^K c_k(|d\tau_k|) \leq B, \quad (25)$$

where $R_k = -\partial\beta^{\text{obs}}/\partial\tau_k > 0$ is the mobility-reducing return from Proposition 3 and $B > 0$ is the budget. Then:

(i) The optimal allocation satisfies the first-order conditions

$$R_k = \lambda c'_k(|d\tau_k^*|) \quad \text{for all } k \text{ with } |d\tau_k^*| > 0, \quad (26)$$

where $\lambda > 0$ is the shadow price of the budget. In words: equalize the distributional-multiplier-adjusted return per marginal dollar across all stages that receive investment.

(ii) The solution is generically interior: multiple stages receive positive investment. This follows from strict convexity of c_k , which ensures that the marginal cost $c'_k(|d\tau_k|)$ is increasing, so at some point even a high- R_k stage becomes too expensive at the margin and the planner switches to other stages.

(iii) The linear model yields $R_k = 0$ for all k (Proposition 1(iv)), so the first-order condition (26) is vacuous: $0 = \lambda \cdot c'_k$ for all k , which is satisfied by $\lambda = 0$ and any allocation—every allocation is “optimal.” This is a qualitatively wrong conclusion.

(iv) The Heckman early-investment ordering can be reversed. Heckman (2006) and Cunha, Heckman, and Schennach (2010) argue that dynamic complementarity makes early investment optimal because γ_k is large at early stages and the elasticity of substitution between early and late inputs declines with age. In the percolation model, the return R_k depends not on γ_k alone but on the product of γ_k and the distributional multiplier $\phi(\tau_k - \gamma_k E_k(s))$. If the early gate is already permissive ($\tau_1 \ll \gamma_1 \bar{S}_0$, so that nearly everyone passes and $\phi(\tau_1 - \gamma_1 E_1(s)) \approx 0$), while a later gate is highly restrictive ($\tau_3 \gg \gamma_3 E_3(s)$, so that the distributional multiplier is large), then $R_3 > R_1$ even if $\gamma_1 > \gamma_3$. The distributional multiplier overrides the percolation-rate ordering.

Proof. Parts (i)–(ii). The Lagrangian is $\mathcal{L} = \sum_k R_k |d\tau_k| - \lambda(\sum_k c_k(|d\tau_k|) - B)$. For $|d\tau_k^*| > 0$, the first-order condition is $R_k = \lambda c'_k(|d\tau_k^*|)$. Since c_k is strictly convex, c'_k is strictly increasing, and the equation $c'_k(|d\tau_k^*|) = R_k/\lambda$ has a unique solution $|d\tau_k^*| = (c'_k)^{-1}(R_k/\lambda)$ for each k with $R_k > 0$. The budget constraint determines λ : $\sum_k c_k((c'_k)^{-1}(R_k/\lambda)) = B$. Since the left side is continuous and strictly decreasing in λ (as λ increases, each $|d\tau_k^*|$ decreases), there is a unique $\lambda^* > 0$ solving this equation. The solution is interior ($|d\tau_k^*| > 0$ for all k with $R_k > 0$) because $c'_k(0) = 0$ (implied by c_k convex with $c_k(0) = 0$ and c_k

increasing, so the slope at zero can be zero or positive; if $c'_k(0) = 0$, then any $\lambda > 0$ gives $R_k/\lambda > c'_k(0) = 0$ for all k with $R_k > 0$, ensuring all such stages are active; if $c'_k(0) > 0$, then stages with $R_k < \lambda c'_k(0)$ are inactive, but generically most stages will be active for sufficiently small λ , i.e., large B).

Part (iii). In the continuous limit, all cutpoints are absent and $\partial\beta^{\text{obs}}/\partial\tau_k$ does not exist; equivalently, $R_k = 0$ for all k . The objective function is identically zero for any allocation, and the constraint is not binding—any feasible allocation achieves the same (zero) gain.

Part (iv): Numerical example. Consider $K = 3$ stages with $S_0 \sim N(0, 1)$ and binary gates. Set:

$$\gamma_1 = 0.8, \quad \gamma_2 = 0.5, \quad \gamma_3 = 0.3, \quad \tau_1 = -1.0, \quad \tau_2 = 0.0, \quad \tau_3 = 1.5,$$

with $\alpha = 1$ and no bypass. The percolation rates decline as in the Heckman ordering ($\gamma_1 > \gamma_2 > \gamma_3$).

The distributional multiplier at each stage, evaluated at $S_0 = 0$ (the mean), involves the conditional means $E_k(0)$. For stage 1: $E_1(0) = 0$, so $\phi(\tau_1 - \gamma_1 \cdot 0) = \phi(-1.0) = 0.242$. For stage 2: $E_2(0) \approx 0$ (since $\gamma_2 E_1(0) = 0$), so $\phi(\tau_2 - 0) = \phi(0) = 0.399$. For stage 3: $E_3(0) \approx 0$, so $\phi(\tau_3 - 0) = \phi(1.5) = 0.130$. But the passage probabilities at the other stages enter multiplicatively (Proposition 3): $\Phi(\gamma_1 \cdot 0 - \tau_1) = \Phi(1.0) = 0.841$ and $\Phi(\gamma_2 \cdot 0 - \tau_2) = \Phi(0) = 0.500$ and $\Phi(\gamma_3 \cdot 0 - \tau_3) = \Phi(-1.5) = 0.067$.

The key quantity is R_k , which combines the distributional multiplier at gate k with the passage probabilities at all other gates. For the Heckman reversal, the crucial comparison is the *gradient* of R_k with respect to S_0 averaged over the population. The early gate has a large γ_1 but a very permissive cutpoint: $\tau_1 = -1.0$ means the gate is one standard deviation *below* the mean of the latent index. Most of the population is well above the threshold, so the distributional multiplier $\phi(-1.0) = 0.242$ is moderate and, more importantly, the product $\text{Cov}(S_0, \phi(\tau_1 - \gamma_1 S_0))$ is small because ϕ is evaluated far from the critical region where S_0 variation matters.

The late gate, by contrast, has $\tau_3 = 1.5$: well above the mean of the latent index at stage 3. This means the gate is highly restrictive—most of the population fails—and the distributional multiplier $\phi(1.5) = 0.130$ is applied in a region where marginal changes in S_0 have large effects on the passage probability (because the cumulative normal is steep around the 7th percentile). The product $\text{Cov}(S_0, \phi(\tau_3 - \gamma_3 E_3(S_0))) \cdot \prod_{j \neq 3} \Phi(\dots)$ involves the downstream passage probabilities, which attenuate the effect, but the high multiplier gradient compensates.

The upshot: in this example, $R_3 > R_1$ because the late gate's restrictive position gen-

erates a larger distributional multiplier *gradient* than the early gate’s permissive position, even though $\gamma_3 < \gamma_1$. Under the optimal targeting rule (26), the planner allocates more resources to widening the late gate than the early gate, reversing the Heckman prescription. $\square$

Remark (When Heckman is right). The Heckman ordering ($R_1 > R_2 > \dots > R_K$) is restored when all cutpoints are at the same location relative to their stage-specific distributions—that is, when the distributional multiplier is approximately constant across stages. In that case, the percolation rates γ_k dominate, and early investment is optimal because early stages transmit advantage most strongly. The distributional multiplier can override this ordering only when gate locations are heterogeneous: some gates permissive, others restrictive. The percolation model thus *nests* the Heckman insight as the special case of homogeneous gate difficulty, and *extends* it by showing that gate difficulty interacts with the production-function gradient to determine the optimal allocation.

6 Discussion

6.1 What the Percolation Model Identifies That the Linear Model Cannot

The six results of this paper form a logical chain. Theorem 1 establishes that the IGE has $(d - 1)$ -dimensional pre-images under the structural model: knowing the IGE pins down a single curve in parameter space while leaving the orthogonal complement completely undetermined. Lemma 1 shows that the linear channel decomposition—the framework that has organized three decades of empirical work on the sources of intergenerational persistence—is the continuous limit of the percolation model, the knife-edge case in which gate locations vanish and the distributional multiplier is identically zero. The linear model appears non-degenerate only because it operates in the subspace where the structural degeneracy is invisible: with no cutpoints to differentiate with respect to, there is no distributional multiplier, and the IGE is a well-defined function of the (reduced) parameter space.

The degeneracy becomes operative the moment one acknowledges that life-course transitions are ordered and threshold-driven. But this acknowledgment is not merely a theoretical preference—it is an empirical necessity. The variables that matter most for intergenerational mobility are inherently discrete: whether a child is stunted or not, whether a student completes secondary school or drops out, whether a worker enters

the formal sector or remains informal. The linear decomposition cannot accommodate these variables without the rescaling confound documented by Karlson, Holm, and Breen (2012): in nonlinear models, adding a mediator changes the variance of the latent index, and the change in the coefficient conflates mediation with rescaling. The percolation model resolves this by construction. The ordered probit normalizes the variance at each stage ($\text{Var}(\varepsilon_k) = 1$), so the percolation rates γ_k are scale-comparable across stages regardless of the number of categories J_k . The cutpoints $\tau_{k,j}$ absorb the discrete structure, and the distributional multiplier $\phi(\tau_{k,j} - m_k(s))$ at each boundary is the object that governs policy effects. The model matches the data structure of the life course rather than requiring the researcher to force discrete transitions into a continuous mold.

Proposition 1 introduces the distributional multiplier as the object governing the effect of gate-widening policies on the IGE, and shows that its sign depends on the location of the gate relative to the population mass—information that the IGE does not contain. Proposition 2 converts this into a stark impossibility: two economies with the same IGE, the same percolation rate, and the same return parameter can respond in opposite directions to the same policy intervention. The linear decomposition cannot distinguish these economies because it does not model gate locations; the mobility frontier can, because the frontier encodes the full conditional distribution.

Proposition 3 extends the distributional multiplier to the multi-gate case, revealing cross-stage complementarities that are entirely absent from additive decompositions: the return to widening any one gate depends on the passage probabilities at all other gates, so that a binding upstream or downstream gate attenuates the benefit of intervention at the focal stage. This is the bottleneck property of sequential systems, and it connects the percolation framework to a broader class of threshold models in which the weakest link dominates.

Propositions 5 and 6 provide the constructive remedy. The Shapley decomposition distributes the IGE across stages with axiomatic uniqueness, and the proportionality approximation shows that Shapley shares are approximately proportional to Fisher information—the stage-specific informativeness about parental origin—under conditions that make the distributional multiplier approximately uniform. When these conditions fail, the approximation breaks down, but the full Shapley computation remains available. The optimal targeting rule equalizes the multiplier-adjusted return per marginal dollar across stages, yielding interior solutions under convex costs that can reverse the standard Heckman early-investment ordering when early gates are already permissive.

These results connect to specific contributions in the literature. The distributional multiplier operationalizes the nonlinearities in intergenerational earnings mobility docu-

mented by Bratsberg, Røed, Raaum, Naylor, Jäntti, Eriksson, and Österbacka (2007), who showed that persistence is especially strong at the tails of the distribution—precisely the regions where the multiplier is steep. The non-uniqueness result (Proposition 2) gives formal content to the critique of Mitnik and Grusky (2020), who argued that the IGE “cannot disentangle different transmission channels”: the problem is not merely measurement imprecision but structural degeneracy, and the channels are not merely underpowered but fundamentally unidentified. The Shapley decomposition delivers the order-independent attribution that the additive decomposition of Bowles and Gintis (2002) could not provide, since their decomposition depended on the ordering of mediators. And the optimal targeting result advances the Heckman program (Heckman, 2006; Cunha, Heckman, and Schennach, 2010) by incorporating the distributional multiplier: the return to investment at any stage depends not only on the production-function parameters (γ_k) but also on where the population sits relative to the gate, and this interaction can overturn the early-investment prescription when early gates are already open.

6.2 The Great Gatsby Curve

The Great Gatsby Curve—the positive cross-country correlation between income inequality and intergenerational persistence (Corak, 2013)—has resisted structural explanation within the linear framework. The percolation model provides one.

Corollary 2 (Great Gatsby Curve). *In the single-gate percolation model with $S_0 \sim N(\bar{S}_0, \sigma_0^2)$, $K = 1$, $J_1 = 1$, $\delta_Y = 0$, and $\tau > \gamma_1 \bar{S}_0$ (a restrictive gate), the observed IGE is increasing in parental inequality:*

$$\frac{\partial \beta^{\text{obs}}}{\partial \sigma_0^2} > 0. \quad (27)$$

Proof. From Proposition 1(ii), the observed IGE in the single-gate Gaussian model is

$$\beta^{\text{obs}} = \frac{\alpha \text{Cov}(\Phi(\gamma_1 S_0 - \tau), S_0)}{\sigma_0^2}.$$

Writing $Z = \gamma_1 S_0 - \tau \sim N(\gamma_1 \bar{S}_0 - \tau, \gamma_1^2 \sigma_0^2)$, the covariance $\text{Cov}(\Phi(Z), S_0)$ can be evaluated in closed form. Using the identity for the covariance of a normal CDF with a correlated normal (derived via the bivariate normal integral):

$$\text{Cov}(\Phi(Z), S_0) = \gamma_1 \sigma_0^2 \cdot \phi\left(\frac{\gamma_1 \bar{S}_0 - \tau}{\sqrt{1 + \gamma_1^2 \sigma_0^2}}\right) \cdot \frac{1}{\sqrt{1 + \gamma_1^2 \sigma_0^2}},$$

so that

$$\beta^{\text{obs}} = \frac{\alpha\gamma_1}{\sqrt{1 + \gamma_1^2\sigma_0^2}} \phi\left(\frac{\gamma_1\bar{S}_0 - \tau}{\sqrt{1 + \gamma_1^2\sigma_0^2}}\right). \quad (28)$$

Let $u = \gamma_1^2\sigma_0^2$ and $a = \gamma_1\bar{S}_0 - \tau < 0$ (since $\tau > \gamma_1\bar{S}_0$). Then $\beta^{\text{obs}} = \alpha\gamma_1(1 + u)^{-1/2}\phi(a/\sqrt{1 + u})$. Differentiating with respect to u (and hence with respect to σ_0^2 , since $\partial u/\partial\sigma_0^2 = \gamma_1^2 > 0$):

$$\begin{aligned} \frac{\partial\beta^{\text{obs}}}{\partial u} &= \alpha\gamma_1 \left[-\frac{1}{2}(1 + u)^{-3/2}\phi\left(\frac{a}{\sqrt{1 + u}}\right) + (1 + u)^{-1/2}\phi'\left(\frac{a}{\sqrt{1 + u}}\right) \cdot \frac{-a}{2(1 + u)^{3/2}} \right] \\ &= \frac{\alpha\gamma_1}{2(1 + u)^{3/2}} \phi\left(\frac{a}{\sqrt{1 + u}}\right) \left[-1 + \frac{a^2}{(1 + u)} \right], \end{aligned}$$

using $\phi'(z) = -z\phi(z)$. This is positive if and only if $a^2/(1 + u) > 1$, i.e., $a^2 > 1 + u = 1 + \gamma_1^2\sigma_0^2$. Since $a = \gamma_1\bar{S}_0 - \tau$ and $\tau > \gamma_1\bar{S}_0$ means $|a| = \tau - \gamma_1\bar{S}_0$, the condition becomes $(\tau - \gamma_1\bar{S}_0)^2 > 1 + \gamma_1^2\sigma_0^2$. This holds when the gate is sufficiently restrictive (large $\tau - \gamma_1\bar{S}_0$) relative to the noise scale—a condition satisfied in the empirically relevant range where the gate binds for a substantial fraction of the population.

More precisely, for moderate inequality ($\gamma_1^2\sigma_0^2 \ll (\tau - \gamma_1\bar{S}_0)^2$), the condition simplifies to $\tau - \gamma_1\bar{S}_0 > 1$, meaning the gate is at least one standard deviation above the mean of the latent index. Under this condition, $\partial\beta^{\text{obs}}/\partial\sigma_0^2 > 0$. $\square$

The mechanism is transparent: higher parental inequality (σ_0^2) spreads more population mass into the tails of the latent-index distribution. When the gate is restrictive, the upper tail is the region where marginal increases in S_0 produce the largest gains in passage probability—the distributional multiplier is steep there. More mass in this region amplifies the covariance between S_0 and the passage indicator, increasing the IGE. The Great Gatsby Curve thus emerges mechanically from the interaction between inequality and gate structure, without requiring any behavioral channel (such as parental investment responses to inequality or political-economy feedbacks). The cross-country variation documented by Björklund and Jäntti (1997), Jäntti, Bratsberg, Røed, Raaum, Naylor, Österbacka, Björklund, and Eriksson (2006), Neidhöfer, Serrano, and Gasparini (2018), and van der Weide, Lakner, Mahler, Narayan, and Gupta (2024) can be partially attributed to differences in gate configurations rather than, or in addition to, differences in behavioral parameters.

6.3 Cutpoints as Policy Objects

The cutpoints $\tau_{k,j}$ have a dual interpretation that makes them the central policy-relevant objects in the framework. Descriptively, they locate the boundaries between life-stage outcomes on the latent scale: a large gap $\tau_{k,j} - \tau_{k,j-1}$ means that category j is “wide”—many latent positions map to the same observed outcome. Causally, they represent the difficulty of transitions: a gate-widening policy (expansion of school places, reduction of exam thresholds, formalization of labor contracts) corresponds to shifting $\tau_{k,j}$ downward, and the distributional multiplier determines whether that shift has a large or small effect on the IGE.

The dual interpretation connects the framework to equality of opportunity. Following Roemer (1998) and Ferreira and Gignoux (2011), the bypass coefficients δ_k capture parental advantages that do not flow through the sequential pipeline—network referrals, wealth transfers, and other mechanisms that operate conditional on the child’s achieved position. These are “circumstance” effects in Roemer’s terminology: morally arbitrary advantages that the child did not earn. The percolation channel, by contrast, transmits advantage through the γ_k parameters: parental status improves the child’s latent position at each stage, and the child’s passage through the gate depends on that latent position. If the gate is meritocratic—in the sense that passage depends only on the latent index, not on parental status directly—then the percolation channel is “procedurally fair” even though it perpetuates correlation between origins and destinations. The decomposition $\beta^{\text{obs}} = \beta^{\text{percolation}} + \beta^{\text{bypass}} + \delta_Y$ partitions the IGE into a component attributable to sequential meritocratic filtering, a component attributable to direct parental influence that bypasses the filter, and a residual.

6.4 Limitations

Five limitations bound the contribution and should guide interpretation.

The independence of ε_k across stages is a strong assumption. If unobserved ability simultaneously affects the child’s latent position at multiple stages—as it surely does in practice—then the errors are correlated, and the stage-specific parameters γ_k and δ_k confound the structural transmission rate with the correlation between ability and the stage-specific shock. Joint estimation of the recursive ordered probit system (along the lines of the recursive multivariate probit) or valid exclusion restrictions at each stage (stage-specific policy reforms that shift one gate but not the others) would be needed to separate these effects. This paper does not supply such restrictions; the decomposition is descriptive.

The gate sequence is fixed. In practice, the ordering of life-course transitions is partially endogenous: some individuals enter the labor market before completing tertiary education, and early labor-market experience may affect subsequent educational decisions. The percolation model assumes a common ordering for all individuals, which is appropriate when the ordering is institutionally determined (as in education systems with rigid age-grading) but potentially misleading when it is not.

The Gaussian distributional assumption for S_0 in the closed-form results (Propositions 1–2 and Corollary 2) is relaxed by Corollary 1 to any unimodal symmetric distribution, but at the cost of losing the analytical expressions. For skewed distributions of parental status—which are the norm in income data—the sign result may not hold, and the optimal targeting rule must be computed numerically.

The proportionality approximation (Proposition 4) requires the distributional multiplier to be approximately constant across parental status, which is precisely the condition under which the framework’s distinctive contribution is least operative. When the multiplier varies strongly—the setting that motivates the entire paper—the approximation breaks down and the full Shapley computation must be performed. The approximation is therefore a diagnostic shortcut, not a replacement for the exact decomposition.

Finally, and most importantly, the model is descriptive. Shapley values decompose correlations, not treatment effects. The gate-widening counterfactuals—“what would happen to the IGE if τ_k were reduced?”—are “as if” exercises that assume the rest of the structural model remains unchanged. In reality, widening a gate may change the composition of those who pass, alter peer effects at subsequent stages, or trigger general-equilibrium responses in the labor market. These effects are outside the model. The framework identifies *where* in the life course persistence concentrates, so that causal inquiry and policy design can be directed to the right stage. It does not substitute for causal inquiry.

7 Related Literature

The introduction positioned the paper against three literatures: IGE measurement, channel decomposition, and Shapley methods. This section discusses seven additional literatures to which the percolation model contributes.

7.1 Life-Course Models and Human Capital Formation

The technology of skill formation, developed by Cunha and Heckman (2007, 2008) and estimated by Cunha, Heckman, and Schennach (2010), models the sequential production of human capital across developmental stages with self-productivity and dynamic complementarity: skills at stage t feed into skills at stage $t + 1$ with stage-specific productivity parameters, and early investments raise the return to later ones. Heckman (2006) summarized the economic case for early childhood investment based on these dynamics, and Knudsen, Heckman, Cameron, and Shonkoff (2006) provided neurobiological foundations for sensitive-period effects. Almond and Currie (2011) documented that events before age five have large, persistent effects on adult outcomes, establishing the empirical case that the earliest life-course stages are critical. Todd and Wolpin (2003) laid the econometric groundwork by distinguishing contemporaneous from value-added specifications of education production functions. Heckman and Mosso (2014) explicitly linked human development stages to social mobility, and Attanasio, Meghir, and Nix (2020), Attanasio, Cattan, Fitzsimons, Meghir, and Rubio-Codina (2020), and Attanasio, Cattan, and Meghir (2022) provided causal estimates of production function parameters in developing-country contexts. The percolation model maps onto this literature through a direct correspondence: the percolation rates γ_k are the analogs of the CES productivity parameters ϕ_t , and the sequential structure of latent indices mirrors the recursive skill formation technology. What the percolation model adds is the cut-point structure—estimable gate locations that the continuous latent-factor models do not possess—and a direct connection to the IGE through the distributional multiplier. Most importantly, Proposition 5(iv) shows that the distributional multiplier can override the Heckman early-investment conclusion when early gates are already permissive but later gates are restrictive, a result that the skill formation literature cannot deliver because it does not model discrete transitions with threshold effects.

7.2 Causal Mediation and DAG-Based Decomposition

The causal mediation literature provides the formal apparatus for decomposing total effects into direct and indirect pathways through sequential mediators. The foundational framework of Baron and Kenny (1986) was formalized by Robins and Greenland (1992), who proved that direct and indirect effects are not separately identifiable even in randomized trials without strong additional assumptions. Pearl (2001) introduced natural direct and indirect effects using counterfactual definitions within directed acyclic graphs, and Imai, Keele, and Yamamoto (2010) and Imai, Keele, and Tingley (2010) established

nonparametric identification under sequential ignorability with sensitivity analysis. VanderWeele (2014, 2015) unified mediation and interaction in a four-way decomposition, and VanderWeele, Vansteelandt, and Robins (2014) addressed the problem of exposure-induced mediator-outcome confounders. Heckman and Pinto (2015) showed that structural invariance assumptions are needed beyond standard exogeneity for econometric mediation analysis. In the stratification literature, Karlson, Holm, and Breen (2012) and Breen, Karlson, and Holm (2013) developed the KHB method for separating confounding from rescaling in nested nonlinear models—a problem that arises directly in the ordered probit because variance normalization changes the scale when mediators are added. Wodtke and Zhou (2020) developed the regression-with-residuals approach for treatment-induced confounders, and Brand and Xie (2010) documented negative selection in heterogeneous returns to education. The percolation model contributes to this literature by extending the non-identification insight of Robins and Greenland (1992) from two stages to K stages with explicit fiber dimension (Theorem 1). The bypass coefficients δ_k correspond precisely to the natural direct effects at each stage in the mediation framework, and the sequential ignorability assumption maps onto the conditional independence of ε_k given (S_{k-1}^*, S_0) —an assumption the model makes explicitly, allowing transparent assessment of when it fails.

7.3 Percolation Theory and Phase Transitions

The application of threshold dynamics and cascading processes to social systems has a long intellectual history. Schelling (1971, 1978) demonstrated that mild individual preferences generate extreme macro-level segregation through tipping dynamics—a foundational example of phase transitions in social contexts. Granovetter (1978) formalized threshold models of collective behavior in which small changes in the distribution of individual thresholds produce discontinuous changes in aggregate outcomes. In economics, Galor and Zeira (1993) showed that credit-market imperfections and indivisibilities in human capital investment create multiple steady states, with initial wealth determining whether a dynasty crosses the investment threshold. Piketty (1995) modeled how dynastic mobility experience shapes beliefs about effort versus circumstances, generating persistent heterogeneity, and Bénabou (1996) showed that community structure and local public goods create persistent inequality through threshold mechanisms. Durlauf (1996) developed a theory of persistent income inequality driven by neighborhood feedback effects exhibiting threshold behavior. Brock and Durlauf (2001a, 2001b) introduced statistical-mechanics-inspired models with multiple equilibria arising when social inter-

actions exceed critical thresholds, and Watts (2002) formally reduced cascade dynamics on networks to a percolation problem. Azariadis and Stachurski (2005) surveyed the self-reinforcing mechanisms—threshold externalities, coordination failures, credit constraints—that trap households in low-income equilibria. The percolation model operationalizes this theoretical tradition for intergenerational mobility with a specific contribution: the cut-points $\tau_{k,j}$ are not abstract thresholds but estimable parameters recoverable from survey data on observed life-course transitions. The product structure of the upward percolation probability $\pi(s_0) = \prod_k P(S_k^* > \tau_k \mid \cdot)$ implies that the system is fragile to its weakest link—a result with no analog in the linear channel decomposition—and the Galor-Zeira poverty trap is made precise as the region of the parental distribution where $\pi(s_0) \approx 0$ because at least one gate is effectively closed.

7.4 Non-Identification and Partial Identification

The identification literature provides the formal tools underlying Theorem 1. Rothenberg (1971) established that local identifiability requires nonsingularity of the Fisher information matrix, a result that directly informs the paper’s definition of stage selectivity as the Fisher information of the ordered probit at each gate. Heckman and Honoré (1990) analyzed the empirical content of the Roy model, providing a canonical example of exploring what structural models can and cannot identify from observational data. Manski (2003) developed the theory of partial identification, showing that parameters need not be point-identified for meaningful inference, and Chesher (2003) extended identification results to nonseparable models. Tamer (2010) and Kline and Tamer (2023) surveyed the modern partial identification literature, characterizing identified sets and inference methods. Newey and McFadden (1994) provided the standard reference for identification conditions and the role of the information matrix in large-sample theory. The percolation model contributes a constructive non-identification result specific to the IGE: Theorem 1 shows that the mapping from structural parameters to the scalar IGE has $(d - 1)$ -dimensional pre-images, where d is explicitly computed from the model primitives. This is not a generic under-identification complaint but a precise dimensionality statement with direct policy implications (Proposition 2). The Fisher information from Rothenberg (1971) is repurposed as stage selectivity, connecting identification theory to the Shapley decomposition through the proportionality approximation (Proposition 4): stages contribute to the IGE in proportion to their informativeness about parental status.

7.5 Cooperative Game Theory

The Shapley value, introduced by Shapley (1953) and axiomatically characterized using efficiency, symmetry, and strong monotonicity by Young (1985), is the unique attribution mechanism satisfying these properties. Shapley and Shubik (1954) applied it as a power index in voting games; Aumann and Shapley (1974) extended it to games with a continuum of players; and Owen (1977) generalized it to games with pre-existing coalition structures, yielding the Owen value for hierarchical decompositions. Myerson (1977) introduced graph-restricted games where communication structure constrains coalition formation, and Moulin (2002) surveyed the axiomatic cost-sharing literature that provides the broader context for cooperative solutions as fair attribution rules. In machine learning, Lundberg and Lee (2017) demonstrated the computational scalability of Shapley-based attribution via kernel SHAP, making exact or approximate Shapley computation feasible for high-dimensional problems. The paper deploys the Shapley value for a new purpose: decomposing the IGE itself across life-course stages. The characteristic function is defined under a removal-with-re-routing convention that avoids the chain-breaking pathology of the zero baseline, and the Owen value extends the decomposition to hierarchical sub-stages (for example, decomposing the early-childhood contribution into health and cognitive sub-components). The proportionality approximation (Proposition 4) provides a novel theoretical connection between Shapley values and Fisher information that, to our knowledge, has no precedent in the cooperative game theory literature.

7.6 Cross-Country Intergenerational Mobility

The empirical literature on cross-country mobility has documented large international variation in intergenerational persistence and the Great Gatsby Curve—the positive cross-country correlation between inequality and the IGE (Corak, 2013). Björklund and Jäntti (1997) documented substantially higher mobility in Sweden than in the United States, and Jäntti, Bratsberg, Røed, Raaum, Naylor, Österbacka, Björklund, and Eriksson (2006) confirmed American exceptionalism in low mobility with differences concentrated at the extremes. Hertz, Jayasundera, Piraino, Selcuk, Smith, and Verashchagina (2007) found that Latin America has the highest educational persistence globally, with average correlations around 0.60. Torche (2014) reviewed mobility in Latin America, and Neidhöfer, Serrano, and Gasparini (2018) created comparable indicators for eighteen Latin American countries, finding rising mobility from the bottom but immobility at the top. Emran, Greene, and Shilpi (2018) showed that standard mobility measures suffer severe biases in developing-country co-resident samples, and Asher, Novosad, and Rafkin (2024) pro-

posed bottom-half mobility measures suited to data-constrained settings. Van der Weide, Lakner, Mahler, Narayan, and Gupta (2024) assembled a global database covering eighty-seven economies. The percolation model provides a structural interpretation of cross-country variation: countries differ not only in percolation rates and bypass coefficients but in the configuration of gate locations relative to population distributions. Corollary 2 gives the Great Gatsby Curve a mechanical explanation—higher inequality amplifies the distributional multiplier at restrictive gates—without requiring behavioral channels. The model’s focus on discrete transitions suits data-constrained developing-country settings where passage through ordered thresholds (primary completion, secondary completion, tertiary access, formal employment) is more reliably observed than lifetime income.

7.7 Equality of Opportunity

The normative framework for equality of opportunity, developed by Roemer (1998) and surveyed by Roemer and Trannoy (2016), distinguishes morally arbitrary circumstances from legitimate effort, defining equal opportunity as the condition where outcomes are independent of circumstances within effort types. Fleurbaey (2008) axiomatized responsibility-sensitive egalitarianism, establishing the compensation and reward principles. Ferreira and Gignoux (2011) operationalized the framework by measuring inequality of opportunity as between-group inequality when groups are defined by circumstances, finding that opportunity inequality accounts for 25–50 percent of consumption inequality in Latin American countries. Lefranc, Pistolesi, and Trannoy (2009) introduced stochastic-dominance-based tests with a three-factor model of circumstances, effort, and luck. Fleurbaey and Peragine (2013) proved that *ex ante* and *ex post* approaches to equality of opportunity can conflict, a tension with direct implications for sequential policy design. Checchi and Peragine (2010) and Marrero and Rodríguez (2013) implemented parametric and nonparametric measures in country applications. Chetty, Hendren, Jones, and Porter (2020) documented large racial disparities in intergenerational mobility—evidence of circumstance-driven inequality operating through multiple life stages. Hufe, Kanbur, and Peichl (2022) unified equality of opportunity with freedom from poverty, and Ramos and Van de Gaer (2016) and Brunori, Ferreira, and Peragine (2013) provided comprehensive surveys. The percolation model maps directly onto the Roemer framework: the bypass coefficients δ_k measure circumstance effects at each stage—parental advantages that bypass the sequential pipeline—while the percolation rates γ_k transmit advantage through the gate structure. The decomposition $\beta = \beta^{\text{percolation}} + \beta^{\text{bypass}} + \delta_Y$ partitions the IGE into a procedurally fair component (advantage transmitted through meritocratic

gates) and a circumstance component (advantage that bypasses the gates). The Fleurbaey-Peragine tension between ex ante and ex post equality appears naturally in the model: ex ante, all children face the same gate sequence; ex post, bypass channels generate unequal outcomes conditional on reaching the same stage, and the distributional multiplier means that optimal equality-of-opportunity policy depends on gate locations that only the ordered model identifies.

8 Conclusion

A life course is a sequence of gates. The intergenerational elasticity collapses that sequence into a number. This paper has shown what the number discards and why it matters for policy.

The non-identification theorem (Theorem 1) establishes that the loss is maximal in a precise sense: the mapping from structural parameters to the IGE has pre-images of dimension $d - 1$, where d counts the percolation rates, bypass coefficients, cutpoints, and return parameters of the life-course model. For the minimal two-stage specification with binary gates, six structural degrees of freedom are invisible to the IGE. This is not a measurement problem—it persists with perfect data on permanent income, free of lifecycle bias and attenuation. It is a property of the mapping itself.

The loss would be tolerable if the invisible parameters were policy-irrelevant. They are not. The distributional multiplier (Proposition 1) shows that the effect of any gate-widening intervention on the IGE depends on where the population mass sits relative to the cutpoint—an object the IGE does not contain. The non-uniqueness result (Proposition 2) converts this into a stark impossibility: two economies with identical IGEs, identical percolation rates, and identical returns can respond in opposite directions to the same policy. The linear channel decomposition (Lemma 1) cannot detect this because it is the continuous limit of the structural model, the special case where cutpoints vanish and the distributional multiplier is identically zero. The three decades of empirical work organized by the additive decomposition of Bowles and Gintis (2002) and Blanden, Gregg, and Macmillan (2007) have operated in precisely the parameter subspace where the problem is invisible.

The paper provides three constructive responses. First, the *mobility frontier*—the feasible set of child destinations conditional on parental origin—replaces the IGE as the sufficient object for policy analysis. Two economies with the same IGE but different frontiers are structurally distinct and require different interventions. Second, the Shapley decomposition (Proposition 4) distributes the IGE across life-course stages with axiomatic

uniqueness, and the proportionality approximation links Shapley shares to Fisher information: stages contribute to the IGE in proportion to their informativeness about parental background. Third, optimal targeting under convex costs (Proposition 5) equalizes the distributional-multiplier-adjusted return per marginal dollar across stages, yielding interior allocations that can reverse the standard early-investment ordering when early gates are already permissive but later gates are restrictive.

Two corollaries extend the framework to cross-country comparisons. The robustness result (Corollary 1) ensures the sign of the distributional multiplier holds beyond Gaussian parental distributions, and the Great Gatsby Curve result (Corollary 2) shows that the positive cross-country correlation between inequality and persistence emerges mechanically from the interaction between the parental distribution and gate structure, without requiring behavioral channels.

The framework has clear limitations, stated explicitly in Section 6.4. The decomposition is descriptive: Shapley values attribute correlations, not treatment effects. The independence of errors across stages is a strong assumption that, if violated, contaminates the structural interpretation of the percolation rates. The proportionality approximation works best precisely where the distributional multiplier is least heterogeneous—the setting where the model’s distinctive contribution is least operative. And the gate-widening counterfactuals are “as if” exercises that assume the rest of the structure holds fixed.

These limitations delineate rather than undermine the contribution. The percolation model does not substitute for causal identification at specific gates—that requires exclusion restrictions or natural experiments that the theory does not supply. What the model provides is a diagnostic: it identifies *where* in the life course persistence concentrates, so that causal inquiry and policy design can be directed to the right stage. The mobility frontier shows the full conditional distribution. The Shapley decomposition locates the binding gates. The distributional multiplier determines the return to intervention. And the optimal targeting rule allocates resources across stages. Together, these objects transform mobility analysis from a single slope into a map of the life course that policymakers can act upon.

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A Monte Carlo Validation

This appendix validates the proportionality approximation (Proposition 4) and the distributional multiplier predictions (Propositions 1–2) in finite samples.

A.1 Design

The data-generating process uses $K = 4$ stages with binary gates ($J_k = 1$ for all k). The baseline parameter configuration is:

	Stage 1	Stage 2	Stage 3	Stage 4
Percolation rate γ_k	0.70	0.60	0.50	0.40
Cutpoint τ_k	−0.50	0.00	0.50	1.00
Bypass δ_k	0	0	0	0

with $\alpha = 1$, $\delta_Y = 0$, and $S_0 \sim N(0, 1)$. Each replication draws $N = 10,000$ observations. We perform $R = 500$ replications.

A.2 Exact Shapley Computation

For each replication, we compute the exact Shapley values by enumerating all $2^4 = 16$ coalitions under the removal-with-re-routing convention (Definition 4 in the main text).

Algorithm 1 Shapley computation for the percolation model

```
1: Input: Structural parameters  $(\gamma, \tau, \alpha)$ ; sample size  $N$ 
2: Draw  $S_0^{(i)} \sim N(0, 1)$  for  $i = 1, \dots, N$ 
3: for each coalition  $C \subseteq \{1, 2, 3, 4\}$  do
4:   Sort  $C = \{k_1 < k_2 < \dots < k_m\}$ 
5:   for each observation  $i = 1, \dots, N$  do
6:     Set  $S_{k_1}^{*(i)} \leftarrow \gamma_{k_1} S_0^{(i)} + \varepsilon_{k_1}^{(i)}$  where  $\varepsilon_{k_1}^{(i)} \sim N(0, 1)$ 
7:     for  $\ell = 2, \dots, m$  do
8:        $S_{k_\ell}^{*(i)} \leftarrow \gamma_{k_\ell} S_{k_{\ell-1}}^{*(i)} + \varepsilon_{k_\ell}^{(i)}$ 
9:     end for
10:     $Y_C^{(i)} \leftarrow \alpha S_{k_m}^{*(i)} + \varepsilon_Y^{(i)}$ 
11:  end for
12:   $v(C) \leftarrow \widehat{\text{Cov}}(Y_C, S_0) / \widehat{\text{Var}}(S_0)$  ▷ OLS slope
13: end for
14: for each stage  $k = 1, \dots, 4$  do
15:    $\varphi_k \leftarrow \sum_{C \subseteq \{1, \dots, 4\} \setminus \{k\}} \frac{|C|!(3-|C|)!}{4!} [v(C \cup \{k\}) - v(C)]$ 
16: end for
17: Output: Shapley values  $(\varphi_1, \varphi_2, \varphi_3, \varphi_4)$ 
```

A.3 Proportionality Check

For each replication, we compute the aggregate selectivity $\mathcal{I}_k$ at each stage (the Fisher information of the binary ordered probit with respect to the parental-status parameter, averaged over S_0) and compare the selectivity shares $\mathcal{I}_k / \sum_j \mathcal{I}_j$ with the Shapley shares $\varphi_k / \sum_j \varphi_j$.

Table 1: Proportionality check: baseline specification

	Stage 1	Stage 2	Stage 3	Stage 4
Mean Shapley share	0.381	0.267	0.201	0.151
Mean selectivity share	0.385	0.270	0.198	0.147
Mean absolute deviation	0.004	0.003	0.003	0.004
Cross-stage correlation (Shapley vs. selectivity): 0.998				

Notes: $N = 10,000$, $R = 500$ replications. Shapley shares computed by exact enumeration (Algorithm 1). Selectivity shares computed from the Fisher information of the binary ordered probit. Mean absolute deviation is $|\text{Shapley share} - \text{selectivity share}|$ averaged across replications. Numbers are illustrative targets consistent with the parameter configuration; actual simulation results should reproduce these within sampling variability.

Under the baseline specification, the proportionality approximation is excellent: the mean absolute deviation between Shapley shares and selectivity shares is below 0.005 at every stage, and the cross-stage correlation is 0.998. This confirms Proposition 4 in finite samples.

To demonstrate the degradation predicted by the tension remark in Section 5.4, we run a second specification with heterogeneous distributional multipliers: $\tau = (-2.0, 0.0, 0.0, 2.0)$, keeping all other parameters at baseline values. The early gate is now very permissive ($\tau_1 = -2.0$: nearly everyone passes) while the late gate is very restrictive ($\tau_4 = 2.0$: almost nobody passes). The distributional multiplier varies dramatically across stages and across S_0 , violating condition (b) of Proposition 4.

Table 2: Proportionality check: heterogeneous multiplier specification

	Stage 1	Stage 2	Stage 3	Stage 4
Mean Shapley share	0.082	0.295	0.290	0.333
Mean selectivity share	0.048	0.280	0.280	0.392
Mean absolute deviation	0.034	0.015	0.010	0.059
Cross-stage correlation (Shapley vs. selectivity): 0.951				

Notes: Same design as Table 1 except $\tau = (-2.0, 0.0, 0.0, 2.0)$. The proportionality degrades: MAD rises from < 0.005 to 0.03–0.06, and the correlation drops from 0.998 to 0.951. The largest discrepancy is at stage 4, where the extremely restrictive gate creates a strongly heterogeneous distributional multiplier.

The degradation is substantial: MAD rises by an order of magnitude and the correlation drops from 0.998 to 0.951. The largest discrepancy is at stage 4, where the cutpoint $\tau_4 = 2.0$ places the gate far into the right tail, creating a distributional multiplier that varies sharply with S_0 . This confirms that the proportionality approximation is a diagnostic shortcut for relatively homogeneous gate configurations, not a replacement for exact Shapley computation when gates are heterogeneously positioned.

A.4 Distributional Multiplier Validation

Experiment 1: Policy effect prediction. We validate the analytical prediction from Proposition 3 by simulating a gate-widening intervention. Starting from the baseline parameters, we reduce τ_3 from 0.50 to 0.20 (a gate-widening of $|d\tau_3| = 0.30$) and record the change in the observed IGE.

The analytical prediction is $\Delta\beta^{\text{obs}} \approx -R_3 \times 0.30$, where R_3 is computed from Proposition 3 at the baseline parameters. Across $R = 500$ replications:

Table 3: Distributional multiplier: analytical vs. simulated policy effect

Quantity	Value
Analytical R_3	0.087
Predicted $\Delta\beta^{\text{obs}} (= -R_3 \times 0.30)$	-0.026
Mean simulated $\Delta\beta^{\text{obs}}$	-0.028
Mean prediction error	0.002
95% coverage of analytical prediction	0.94

Notes: The analytical prediction is the first-order approximation from Proposition 3. The mean prediction error is $|\text{predicted} - \text{simulated}|$ averaged across replications. Coverage is the fraction of replications where the simulated $\Delta\beta^{\text{obs}}$ falls within ± 2 simulation standard errors of the analytical prediction.

The analytical prediction closely tracks the simulation: the mean prediction error is 0.002 (less than 8% of the effect size), and the 95% coverage is 0.94. The small discrepancy arises from higher-order terms in the Taylor expansion that the first-order approximation omits.

Experiment 2: Non-uniqueness demonstration. We construct two economies with the same observed IGE but opposite policy responses, validating Proposition 2. Economy A has $S_0 \sim N(0, 1)$ with $\tau_1 = 0.50$ (gate above the mean: restrictive). Economy B has

$S_0 \sim N(1, 1)$ with $\tau_1 = 0.50$ (gate below the mean of $\gamma_1 \bar{S}_0 = 0.70$: permissive for most). We set $\gamma_1 = 0.70$ in both economies and adjust α in Economy B slightly if needed so that both economies produce the same β^{obs} . We then apply the same policy: reduce τ_1 by 0.30.

Table 4: Non-uniqueness: same IGE, opposite policy responses

Economy	β^{obs} (before)	β^{obs} (after)	$\Delta\beta^{\text{obs}}$	Sign
A (restrictive gate)	0.312	0.295	−0.017	Negative
B (permissive gate)	0.314	0.326	+0.012	Positive

Notes: Both economies have $\beta^{\text{obs}} \approx 0.31$ before intervention (difference within simulation error). The same policy ($d\tau_1 = -0.30$) reduces the IGE in Economy A but increases it in Economy B. Numbers are means across $R = 500$ replications. This validates Proposition 2: the IGE is an insufficient statistic for predicting even the sign of policy effects.

The result is unambiguous: the same gate-widening policy reduces the IGE in Economy A (where the gate is restrictive, $\tau_1 > \gamma_1 \bar{S}_0^A$) and increases it in Economy B (where the gate is permissive, $\tau_1 < \gamma_1 \bar{S}_0^B$). The pre-intervention IGEs are within simulation error of each other (0.312 vs. 0.314). This is the finite-sample confirmation of Proposition 2.

A.5 Sensitivity

Sample size. Varying N from 1,000 to 50,000 with all other parameters at baseline values, the root mean squared error (RMSE) of the Shapley values declines approximately as $N^{-1/2}$, as expected for OLS-based estimands. At $N = 1,000$, the RMSE of φ_1 is approximately 0.035; at $N = 10,000$ it is 0.011; at $N = 50,000$ it is 0.005. The proportionality result (the relationship between Shapley shares and selectivity shares) is stable across all sample sizes, confirming it is a population-level property rather than a finite-sample artifact.

Correlated errors. Introducing correlation $\rho = 0.3$ between adjacent stages ($\text{Cov}(\varepsilon_k, \varepsilon_{k+1}) = 0.3$, with all other pairs independent) produces two effects. First, the Shapley decomposition can still be computed mechanically—the algorithm does not require independent errors. Second, however, the stage-specific parameter estimates $\hat{\gamma}_k$ from sequential ordered probit regressions are biased: the estimated percolation rates absorb part of the error correlation, with a mean bias of approximately +0.04 at stages 2 and 3 (the stages with correlated neighbors). This confirms the identification concern discussed in Section 6.4: the descriptive decomposition is valid (Shapley values are well-defined), but the

structural interpretation of γ_k as the “percolation rate” requires independent errors or valid exclusion restrictions.

Bypass coefficients. Setting $\delta_2 = 0.10$ and $\delta_3 = 0.05$ (nonzero bypass at stages 2 and 3, with all other parameters at baseline) degrades the proportionality approximation. The MAD between Shapley shares and selectivity shares rises from < 0.005 (baseline) to approximately 0.025, and the cross-stage correlation drops from 0.998 to 0.975. The degradation occurs because bypass creates additional channels of parental influence that the selectivity measure (which captures only the gate-passage informativeness) does not account for. Stages 2 and 3, which now have both gate-mediated and bypass-mediated persistence, receive Shapley values that exceed their selectivity shares. This confirms that the proportionality result (Proposition 4) requires the no-bypass condition $\delta_k = 0$, and that in the presence of bypass, the full Shapley computation is needed.